Cluster Computing (2026) 29:920
https://doi.org/10.1007/s10586-026-06730-5

Machine learning-based cloud IDS: a systematic literature review and MITRE ATT&CK mapping

Syed Waqas Ali[1] · Ibrar Ali Shah[1] · Farzana Zahid[2] · Daniyal Munir[3] · Ikram Syed[4] · Md Akbar Hossain[5]

Received: 5 March 2026 / Revised: 5 October 2026 / Accepted: 11 October 2026
© The Author(s) 2026

Abstract

Cloud computing has become the backbone of modern digital transformation, enabling scalable, flexible, and cost-efficient resource provisioning across distributed infrastructures. However, the dynamic and shared environment in cloud computing is vulnerable and exposed to security threats and potential intrusions. An intrusion detection system (IDS) is a key defensive system that detects such attacks and allows them to be blocked. The advancement of machine learning (ML) has significantly transformed several domains, including IDS. To analyze how ML is reshaping cloud security, we conducted a systematic literature review (SLR) with an explicit focus on how attacks studied in the literature align with the MITRE ATT&CK Cloud Matrix. The SLR follows the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework to systematically screen and examine publications from 2020 to 2025. We synthesized and evaluated these technical contributions based on the proposed taxonomy by classifying the studies into three major perspectives to better analyze trends across IDS architectures: the Performance Optimization perspective, Learning and Adaptive perspective, and Environment Aware perspective. Moreover, to provide deeper insights into this evolving landscape in cloud environments, we mapped each attack addressed by existing studies to the state-of-the-art Tactics, Techniques, and Common Knowledge of the MITRE ATT&CK Cloud Matrix, showing that the majority of cloud-specific attack vectors remain largely unaddressed in current cloud-based IDS evaluations. These findings provide insights into the current state of ML-based cloud IDS research and help us identify critical gaps. This work serves as a comprehensive reference for researchers and practitioners aiming to understand the landscape of ML-based IDS in cloud environments.

Keywords Cloud computing · Intrusion detection system · MITRE ATT& CK cloud matrix · Attacks

Syed Waqas Ali, Ibrar Ali Shah, Farzana Zahid, Daniyal Munir, Ikram Syed and Md Akbar Hossain contributed equally to this work.

✉ Ibrar Ali Shah
ibrar@uetmardan.edu.pk

✉ Farzana Zahid
farzana.zahid@waikato.ac.nz

Syed Waqas Ali
waqas.ali@uajk.edu.pk

Daniyal Munir
daniyal.munir@rptu.de

Ikram Syed
ikram@hufs.ac.kr

Md Akbar Hossain
ahossain@eit.ac.nz

[1] Department of Computer Software Engineering, University of Engineering and Technology, Mardan 2320, Pakistan

[2] Department of Computer Science, University of Waikato, Hamilton 3240, Waikato, New Zealand

[3] WICON, RPTU University, Kaiserslautern 67633, Germany

[4] Department of Information and Communication Engineering, Hankuk University of Foreign Studies, Yongin 17035, South Korea

[5] Eastern Institute of Technology, 238 Queen Street, Auckland 1010, New Zealand

Published online: 30 October 2026

1 Introduction

Cloud computing has transformed the modern information technology (IT) infrastructure by enabling on-demand and scalable access to computing resources and services over the Internet [1]. It shifts organizational spending from upfront capital investment in physical infrastructure to flexible, usage-based operational costs, thereby improving operational flexibility and cost-effectiveness [2]. As cloud technologies continue to advance, they are increasingly contributing to the development of cloud-native applications and network architectures, particularly through emerging paradigms such as edge computing, hybrid cloud, and multi-cloud environments, which further improve resource utilization and service delivery [3]. These advancements have enabled organizations to adopt pay-as-you-go models, thereby improving accessibility to substantial computing power and enhancing overall agility [4]. Infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS) are the three primary service models associated with cloud computing [5, 6]. The function of every model has its own unique way of operating in the cloud, with both advantages for operation and with that comes specific security risks. For example, IaaS models have a high risk due to improper configuration. Misconfiguration is one of the most common causes of vulnerabilities within IaaS systems. PaaS models include interfaces and application programming interfaces (APIs) which can be used as points of entry by attackers if they are not properly secured. Data leakage is considered a significant threat to SaaS environments [7].

Additionally, to the above-mentioned model-related security issues, the constantly evolving characteristics of cloud ecosystems create an increased number of complex security threats. Those threats include: ransomware, zero-day attacks, insider threats, supply chain attacks, adversarial attacks, distributed denial of service (DDoS) attacks, advanced persistent threats (APTs), and cloud account hijacking [8, 9]. Such threats result in reputational damage, downtime, and financial loss. In 2024, the global average cost of a data breach reached $4.88 million, the highest figure recorded to date [10]. Other recently reported real-world incidents include the 2024 multi-cloud data warehouse breach of Snowflake [11, 12], an AWS load balancer exploit [13], and cloud resource hijacking [14], which further highlight the increasing exposure of the cloud environment to evolving cyber risks. Therefore, these trends emphasize that ensuring the security of the cloud environment is an ongoing challenge that requires continuous advancements in traditional security mechanisms.

Traditional cloud security mechanisms, such as encryption, web application firewalls, security groups, and access controls, are the current solutions to secure a cloud environment; however, challenges such as improper permissions management, manual interventions, delayed response, evolving threats, and poor key management make them difficult to implement in a dynamic cloud environment [15–17]. To overcome these challenges effectively, intrusion detection systems (IDS) play a crucial defensive role by continuously analyzing and automatically detecting malicious activities [18–22].

An IDS monitors network traffic to automatically detect malicious activity, and ML-based IDS can detect attacks such as DDoS in real time on cloud platforms [23]. Advances in machine learning (ML) and deep learning (DL) have greatly enhanced IDS solutions over traditional statistical methods. Both ML and DL enable IDS solutions to adaptively detect existing as well as emerging threats using learned representations of complex network behavior, thus providing a much greater detection capability than statistical methods [24–26]. Although significant advancements have been made in cloud-based IDS, several issues persist, such as low accuracy, poor generalizability, and a high rate of false alarms [27]. Despite significant progress and a range of focused studies, the current literature lacks comprehensive insight into how ML-based IDS are systematically designed, deployed, and optimized within cloud computing environments. Moreover, a comprehensive mapping of the techniques and tactics used by attackers to target cloud systems remains largely unexplored. **Please note that the term ML-based IDS also refers to DL-based IDS in this study**.

To overcome these gaps, *this study presents a Systematic Literature Review (SLR)* [28] *to identify and analyze current ML-based IDS approaches for cloud environments by exploring the following research questions:*

- RQ1: What is the current landscape of existing ML-based intrusion detection systems in the context of cloud computing?
  - RQ1.1: What are the main categories of ML-based cloud IDS explored in the literature?
  - RQ1.2: What types of ML algorithms, approaches, and security attacks have been considered in the selected primary studies?
  - RQ1.3: How has the applicability of existing ML-based cloud IDS been assessed?
- RQ2: With reference to RQ1, to what extent do the existing ML-based IDS solutions meet real-world cloud security requirements?

Table 1 A comprehensive comparison of existing reviews with our work

Reference	Perspectives								Other aspects		
	Performance Optimization		**Learning and Adaptive**				**Environment Aware**		**Proposed Hierarchy**	**Contemporary Evaluation**	**MITRE Mapping**
	Nature-Inspired IDS	Lightweight IDS	Federated Learning IDS	Anomaly-based IDS	Traditional ML-based IDS	Adversarial Resilient IDS	Hypervisor based IDS	Multi-cloud IDS			
[16]	✓	✗	✗	✓	✓	✗	✓	✗	✓	✗	✗
[19]	✓	✗	✗	✓	✓	✗	✓	✗	✗	✗	✗
[20]	✗	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗
[21]	✓	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗
[22]	✗	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗
[26]	✓	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗
[29]	✓	✗	✗	✓	✓	✗	✓	✗	✓	✓	✗
[30]	✓	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗
[31]	✗	✗	✓	✓	✓	✗	✗	✗	✓	✓	✗
Our Work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Symbols: ✓=Addressed, ✗=Not Addressed

Bold text in the tables is used for formatting purposes only, including to highlight table headings, category names, and "Our Work" in Table 1. It does not indicate statistical significance or the best-performing results

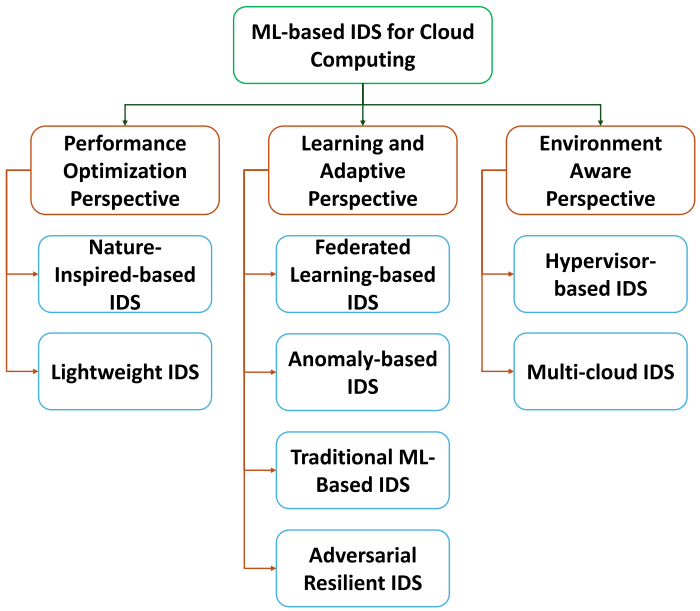


Fig. 1 Proposed taxonomy for organizing the reviewed studies on ML-based cloud IDS

- RQ2.1: How are the identified security aspects aligned with the industry standard MITRE ATT&CK Cloud Matrix?
- RQ2.2: What are the current research gaps in cloud-specific ML-based IDS?

Table 1 includes reviews that broadly survey the ML-based cloud IDS literature, while more focused related work is discussed here. Velliyath and Kalaivani [32] review more recent state-of-the-art with an emphasis on adversarial robustness. Ma et al. [33] review cloud-edge federated learning (FL)-based IDS from a model-optimization perspective, and Buyuktanir et al. [34] survey FL-based IDS across domains such as IoT, healthcare, and cybersecurity.

To answer the research questions, 79 primary studies were selected from 1,013 papers following the SLR protocol discussed in Sect. 2. The major findings of SLR are presented by categorizing existing ML-based cloud IDS approaches into three analytical perspectives: performance optimization, learning and adaptive, and environment-aware. These perspectives capture how ML-based IDS solutions are optimized, learned, adapted, or designed for attack detection in a cloud environment. Figure 1 presents the taxonomy developed from this classification and serves as the

organizational framework for the review. The proposed perspectives and subcategories provide the basis for structuring the detailed discussion of the reviewed studies in Sect. 3.

We also conducted a contemporary evaluation of the identified primary studies across each category using ML-based algorithms, types of security attacks detected, utilized datasets, and performance metrics. The analysis of our SLR helps identify research gaps and the most critical areas in cloud computing that require further investigation. *Another major objective of this study* is to bridge the gaps between academic research and real-world cloud security needs by mapping the identified attacks to the MITRE ATT&CK Cloud Matrix [35]. The MITRE ATT&CK Cloud Matrix is a structured approach that provides critical information about attackers' tactics, techniques, and procedures (TTPs) and helps organizations to effectively respond to emerging attacks in their cloud infrastructure. Furthermore, to the best of our knowledge, related work summarized in Table 1 does not provide an industry-aligned cloud security threat representation. Although many review studies have examined ML-based IDS for cloud computing from various viewpoints (e.g., performance optimization, anomaly detection, lightweight architecture, and environment-aware deployment) [16, 19–22, 26, 29–31], there are still some critical gaps. The key limitations of prior work are described in Table 1, and they provide very little overall coverage of the larger cloud IDS space. Additionally, to the best of our knowledge, the studies have not used an industry-recognized threat framework like the MITRE ATT&CK Cloud Matrix to evaluate the completeness of attack coverage. Therefore, it is challenging to determine if the currently proposed ML-based cloud IDS solutions accurately reflect real-world cloud adversarial behavior and identify where significant detection vulnerabilities exist. Motivated by these limitations, this study provides a complete review that incorporates multiple perspectives of IDS, provides a comparative assessment of recent studies on the topic of IDS in clouds, and uses the MITRE ATT&CK Cloud Matrix to systematically map the reported attacks and identify the research gaps and directions for further research.

The *primary contributions* of this study are as follows:

- A systematic, PRISMA-guided literature review of recent (2020–2025) ML-based IDS for cloud computing is presented, shortlisting 79 primary studies from an initial pool of 1,013 records (Sect. 2).
- A structured taxonomy is proposed that classifies existing ML-based cloud IDS research into three analytical perspectives: performance optimization, learning and adaptive, and environment-aware. This classification shows that research is concentrated mainly on traditional ML-based and nature-inspired optimization-based IDS, whereas federated-learning, adversarial-resilient, hypervisor-based, and multi-cloud IDS remain comparatively less explored (Sect. 3).
- A contemporary evaluation of the primary studies is provided across ML/DL techniques, learning paradigms, IDS types, attacks, datasets, and performance metrics. The analysis shows a strong reliance on supervised learning, network-based IDS, and legacy or simulated datasets, indicating limited maturity in cloud-native and host-level detection as well as in generalizable IDS evaluation (Sect. 3).
- The attacks addressed across the reviewed literature are mapped to the MITRE ATT&CK Cloud Matrix tactics and techniques. To the best of our knowledge, this represents one of the first comprehensive alignments of ML-based cloud IDS studies with the MITRE ATT&CK Cloud Matrix. The mapping shows that studied attacks cluster around tactics such as Initial Access, Execution, and Impact, whereas cloud-relevant tactics including Persistence and Collection remain unaddressed (Sect. 4.1).
- Key research gaps and future directions are identified, including the need for cloud-native benchmark datasets, broader coverage of cloud-specific threats, adversarially robust detection models, greater attention to HIDS and multi-cloud environments, and closer alignment between academic IDS evaluations and real-world cloud security requirements (Sect. 4.2).

A comprehensive list of all the abbreviations used throughout this paper is available in Table 2 for ease of reference.

2 Methodology

This section presents the methods and steps used to conduct the SLR. We selected the state-of-the-art PRISMA framework and followed its guidelines to make the review process clear, credible, and reproducible. The details of the process followed using the selective framework are shown in Fig. 2.

2.1 Scope definition and formulation of research questions

The scope of this research is to identify and analyze existing research works on ML-based IDS within the domain of cloud computing, including both cloud-service-specific and general threats. Based on this scope, we formulated the research questions outlined in Sect. 1, confined to studies published from 2020 to 2025. Please note: statistically-based IDS is beyond the scope of this study.

Table 2 List of acronyms

Acronym	Description	Acronym	Description
AdaBoost	Adaptive Boosting	AE	Autoencoder
Anoml.	Anomaly	BiLSTM	Bidirectional LSTM
CNN	Convolutional Neural Network	DAE	Denoising Autoencoder
DL	Deep Learning	DNN	Deep Neural Network
DRL	Deep Reinforcement Learning	DT	Decision Tree
ENIDS	Enhanced Network Intrusion Detection System	FL	Federated Learning
GAN	Generative Adversarial Network	GBM	Gradient Boosting Machine
GRU	Gated Recurrent Unit	HIDS	Host-based Intrusion Detection System
HL	Hybrid Learning	HOIC	High Orbit Ion Cannon
IaaS	Infrastructure as a Service	IDS	Intrusion Detection System
IP Spoof	IP Spoofing	KNN	K-Nearest Neighbor
LightGBM	Light Gradient Boosting Machine	LOIC	Low Orbit Ion Cannon
LR	Logistic Regression	LSTM	Long Short-Term Memory
ML	Machine Learning	MLA	Machine Learning Approach
NB	Naive Bayes	NG	Not Given
NIDS	Network-based Intrusion Detection System	PaaS	Platform as a Service
RF	Random Forest	RL	Reinforcement Learning
RNN	Recurrent Neural Network	SaaS	Software as a Service
SAE	Stacked Autoencoder	SGD	Stochastic Gradient Descent
SL	Supervised Learning	SVM	Support Vector Machine
UL	Unsupervised Learning	VM Escape	Virtual Machine Escape
XGBoost	Extreme Gradient Boosting	XSS	Cross-Site Scripting
BD	Backdoor Attack	DoS	Denial of Service Attack
DDoS	Distributed Denial of Service Attack	Expl.	Exploitation-based Attack
Fuz.	Fuzzing Attack	Fuzzing	Fuzzing-based Attack
SC	Shellcode Attack	Recon.	Reconnaissance Attack
Worms	Worms Attack	Probe	Probing Attack
R2L	Remote to Local Attack	U2R	User to Root Attack
BF	Brute-Force Attack	Bot	Bot Attack
Botnet	Botnet-based Attack	Infil.	Infiltration Attack
Web	Web-based Attack	WebDDoS	Web DDoS Attack
Keylog	Keystroke Logging Attack	Exfiltration	Data Exfiltration Attack
MITM	Man-in-the-Middle Attack	SQLi	SQL Injection Attack
PS	Port Scanning Attack	SCAN	Scanning Attack
WA	Web Application Attack	ZS	Zero-day Attack
SHT	Shellcode-based Threat	GE	Generic Attack
Insider	Insider Threat	CM	Command Injection Attack
Cryptoj.	Cryptojacking Attack	Malware	Malware Attack

To answer RQ1, we conducted a comprehensive SLR, guided by the PRISMA framework, to explore various categories of ML-based IDS, identify the considered security attacks, analyze the utilized datasets, and the performance criteria of the existing cloud-specific IDS. Please note that we only report the performance criteria mentioned in the selected studies; a critical assessment of each criterion is beyond the scope of our study. However, to avoid treating the reported values as directly comparable benchmark results, we interpret the extracted performance metrics in relation to dataset choice and experimental context. This is important because the reported performance of ML-based IDS models can be strongly influenced by dataset characteristics, feature representation, attack-class distribution, class imbalance, preprocessing strategy, and evaluation protocol. Therefore, the accuracy, precision, recall, and F1-score values reported in this review are treated as indicative trends rather than uniform evidence of deployment-level effectiveness. For RQ2, we mapped the attacks detected by the identified ML-based IDS to the industry-accepted MITRE ATT&CK Cloud Matrix, and analyzed the limitations in the current landscape of cloud-specific security.

The SLR also identifies several secondary studies in this area reported in Table 1 [16, 19–22, 26, 29–31]. However, this related work provides limited insights into ML-based cloud IDS, including adversarial attack resilience,

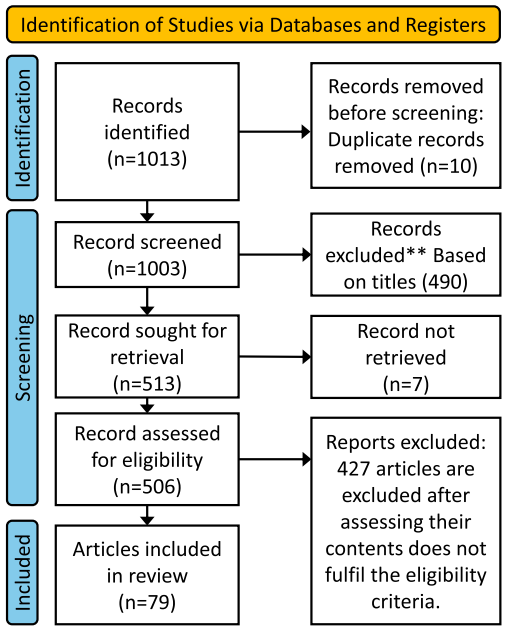


Fig. 2 PRISMA guidelines for extraction and selection of literature

Table 3 Screening stage details

Criteria	Description
Indexing Databases	Web of Science (WoS), IEEE, Springer, ACM, and Scopus were used for indexing.
Keywords Used	"Intrusion detection system" or "IDS"; "Machine learning" or "ML"; "Cloud computing" or "Cloud environment" or "Cloud security".
Search String	(("intrusion detection system" OR "IDS") AND ("machine learning" OR "ML") AND ("deep learning" OR "DL") AND ("security") AND ("cloud computing" OR "cloud environment"))
Screening Focus	Focused on retrieving studies specifically on ML-based IDS in cloud computing environments, ensuring relevance and quality.

lightweight IDS design, FL-based IDS, and multi-cloud IDS. Furthermore, to the best of our knowledge, these existing studies do not provide an explicit mapping of ML-based cloud IDS to industry-recognized frameworks such as the MITRE ATT&CK Cloud Matrix.

2.2 Search (Screening) strategy

A systematic literature review was conducted using the PRISMA framework (Fig. 2) to guide the selection and assessment of primary studies. Five major indexing databases: IEEE Xplore, Springer Link, Scopus, ACM Digital Library, and Web of Science (WoS) were searched. A specific search string was used to identify studies on ML-based intrusion detection systems (IDS) in cloud computing environments, as shown in Table 3. The same search strategy was applied across all databases, and duplicate records were removed before screening. The remaining studies were then screened following the PRISMA guideline using their titles, abstracts, introductions, conclusions, and, where necessary, full texts. To ensure relevance and quality, only peer-reviewed journal and conference articles published in English between 2020 and 2025, with full-text availability, were included in the review. Inclusion and exclusion criteria were also applied to refine the selection of primary studies, as discussed in Sect. 2.3.

2.3 Inclusion and exclusion criteria

The following inclusion criteria (IC) and exclusion criteria (EC) were applied to filter relevant studies from the search results:

Inclusion Criteria (IC):

- IC1: Studies that include ML-based IDS implemented in cloud environments.
- IC2: Articles published from the year 2020-2025.
- IC3: Studies that employ supervised, unsupervised, or hybrid ML learning approaches for IDS.
- IC4: Peer-reviewed journal or conference papers available in full text.
- IC5: Studies discussing algorithms, datasets, or evaluation metrics related to ML-based IDS in cloud computing.**Exclusion Criteria (EC):**

- EC1: IDS-based studies that are not related to cloud computing.
- EC2: Non-machine learning/traditional statistical IDS approaches for cloud environments.
- EC3: Articles before 2020.
- EC4: Duplicate studies or extended versions of previously included papers.
- EC5: Non-peer-reviewed articles, theses, blogs, or survey summaries lacking sufficient technical depth.

2.4 Data extraction and synthesis

The PRISMA framework illustrated in Fig. 2 outlines the systematic selection process adopted in this study to identify relevant studies on ML-based intrusion detection systems in cloud environments. A total of 1013 records were identified using these databases, as shown in Table 3. After removing 10 duplicates and excluding 490 articles based on their titles, 506 full-text articles were selected for further screening. Ultimately, 79 primary studies were included in the final review after excluding 427 studies based on the defined inclusion and exclusion criteria. For each selected study, we synthesized and extracted the following crucial information: research objectives, deployed ML models, datasets utilized,

attacks addressed, evaluation metrics, and performance results. These data were systematically organized and analyzed to identify common trends, methodological patterns, and key insights, offering a comprehensive overview of the current state-of-the-art.

2.5 Quality assessment of this study

To examine the quality of our SLR, a quantitative quality assessment evaluation is performed using quality measures (QC1-QC5), which are based on a three-point scoring system (0, 0.5, 1) as discussed in [36]. The total quality score is the sum of the individual scores, which is quantified as low when the range is ($0.5 \leq$ quality score ≤ 2), medium ($2.5 \leq$ quality score ≤ 3), or high ($3.5 \leq$ quality score ≤ 5).

In addition to evaluating the overall quality of this SLR, we also assessed the methodological quality of all 79 distinct primary studies included in the review. For this purpose, five study-level quality assessment (QA) criteria were used: QA1-clarity of objectives and targeted threats, QA2-methodology description, QA3-dataset reporting, QA4-reported evaluation metrics, and QA5-validation through baselines, comparisons, or multiple datasets. Each criterion was scored using the same three-point scale, where 1 indicates that the criterion is fully satisfied, 0.5 indicates that it is partially satisfied, and 0 indicates that it is not satisfied. The complete study-wise QA results, including criterion-wise scores, total quality score, and quality level for each selected primary study, are reported in Appendix A (Table A1).

The scoring of our study based on these criteria is as follows:

1. QC1. *The inclusion and exclusion criteria* =1 (clearly defined).
2. QC2. *Searching process* =1 (completed through reputed digital libraries).
3. QC3. *Explicit synthesis method* =1 (presented).
4. QC4. *Quality assessment of selected primary studies* =1 (conducted and reported in Appendix A).
5. QC5. *Information about primary studies* =1 (provided).

Based on the defined criteria, our study achieved an overall score of 5 out of 5, reflecting the high quality of the conducted review. Reporting the QA scores of all selected primary studies improves the transparency, traceability, and reproducibility of the review process.

2.6 Conduction reporting (Threats to Validity)

Missing important relevant studies: This SLR was designed to extract a comprehensive set of articles on ML-based IDS for cloud environments, but it is still possible that some relevant studies were missed. To mitigate this risk, we devised a broad search string incorporating multiple synonyms and reviewed introductions, conclusions, and, in some cases, the full texts of the selected primary studies. Additionally, forward and backward snowballing were performed to ensure that all primary studies were included. **Researcher bias:** Potential researcher bias toward peer-reviewed journals and leading conferences may affect the validity of the study; therefore, the review protocol was designed and validated with feedback from domain experts and co-authors, ensuring a consistent and transparent review process. **Number of selected relevant studies:** The final number of studies was determined by the defined scope, research novelty, and degree of overlap among related concepts. As cloud-based IDS often overlaps with related areas such as IoT and edge computing, each study was carefully reviewed in full to ensure relevance. In the case of ambiguity, domain experts were consulted to confirm the inclusion decisions and maintain consistency. **Cross-study performance comparability:** Performance statistics from the chosen primary studies were provided in this assessment. The datasets used, types of attacks that were included, and feature extraction/processing/evaluation approaches each study adopted may differ. The values of accuracy, precision, recall, and/or F1-score results are reported directly from primary studies. Performance metric values are expected to illustrate broad trends and patterns throughout the literature; they are not expected to serve as a basis for direct comparative assessments/benchmarking of the relative merits of different research studies. **Dataset-performance generalisability:** In addition to enabling cross-study comparison, the numerical values representing model performance will also be influenced by the dataset used for that model's performance assessment. Legacy and curated benchmarks (e.g., KDDCup99 and NSL-KDD) typically produce highly accurate models; they are structured and have been commonly employed in a variety of studies, which has resulted in consistent, repeatable use of similar features and attack distributions. The same cannot be said, however, of cloud logs from various platforms and clouds, log files at the container level, multi-cloud traces, and modern cloud-native attacks with varied traffic characteristics and/or non-standardized label assignments. For this reason, high levels of accuracy on benchmark datasets are considered in this review as an indication of within-dataset effectiveness and not as direct indicators of how well those models would perform in real-world deployments. **Researchers' expertise:** Researchers'

expertise can be another factor in determining how accurately they interpret their data when looking at such a large body of research. The authors have a great deal of experience with both cloud security and machine learning; however, maintaining that same level of expertise in each area of sub-speciality is very difficult. To reduce these limitations, constant consultations with domain experts were made available for review and comments. **Dependence on Existing benchmark datasets:** Most of the included primary studies use the same legacy datasets, such as KDDCup99, NSL-KDD, and so on. On the other hand, some studies have chosen to use custom or simulated data. These datasets are still very much in use today as benchmarks for IDS, but they cannot represent well the complexities, the scales, and the types of attacks specifically in cloud computing. Therefore, the trends observed, the attacks covered by this study, and especially the attack coverage represented by the MITRE ATT&CK Cloud Matrix mapping presented in Sect. 4.1, can be seen as being highly dependent upon the type of dataset used in the studies which form the basis of this review. Thus, the limited number of representations of certain tactics (e.g., Persistence and Collection) should be understood through the lens of this issue because most of the benchmark datasets commonly used in the primary studies contain very few, or no representations at all, of cloud-native post-compromise behaviors. Finally, the dependence on custom datasets will affect the generality of the results of the performance metrics reported in this study, and reinforces the necessity for cloud-native benchmark datasets, standardized in terms of their structure and content, to be used as a base for future research and accurately represent the cloud-specific landscape.

3 Current landscape of ML-based IDS in cloud computing

This section addresses the following research question:

RQ1.*What is the current landscape of existing machine learning-based intrusion detection systems in the context of cloud computing?*

3.1 Classification of ML-based cloud IDS in existing literature

In the existing literature, various ML-based cloud IDS have been proposed, with researchers focusing on three perspectives: **performance optimization, learning and adaptation, and environment-aware**. These perspectives and the primary studies that fall under each category are discussed in the following.

3.1.1 Performance optimization perspective

The performance optimization perspective includes primary studies that focus on techniques for improving or enhancing the efficiency and effectiveness of the proposed IDS solutions through feature engineering, reducing computational and memory overheads, and accelerating training and inference. The following subsection discusses the primary studies that fall under each perspective.

Nature-inspired optimization-based IDS: This subsection summarizes the primary studies that integrate machine learning with nature-inspired optimization algorithms to improve the performance of cloud-specific IDS (Table 4). Most studies have focused on swarm intelligence and evolutionary optimization algorithms. For example, chaotic manta ray foraging optimization (CMFO) and multi-task learning DNN (MTL-DNN) algorithms have been used to detect various attacks available in the KDDCup99 and NSL-KDD datasets [37]. Similarly, another approach was proposed in [38] by using adaptive rat swarm optimization (ARSO) with BiLSTM, which yielded 98.2% accuracy. The approach in [39] introduced a four-step horse herd optimization (HHO) with a DL-based intrusion detection approach (HHODL-IDA) pipeline using invasive weed optimization (IWO), BiLSTM, and HHO for hyperparameter tuning, achieving 98.17% accuracy. Deep BiLSTM with marine goal optimization (MGO) achieved an accuracy of 99.2% in [40], whereas honey badger optimization (HBO) improved fuzzy DNN detection in [41], despite generating a high alert volume.

Several studies have explored evolutionary optimization algorithms to enhance the performance of ML-based IDS in cloud environments. For instance, Ghosh et al. [42] utilized the sage grouse mating algorithm (SGMA) for feature selection, achieving 84.5% accuracy with a Decision Tree (DT) classifier. Parkash et al. [43] implemented a cuckoo search with SVM and Naive Bayes and achieved an accuracy of 87.5%. Similarly, Sarkar et al. [44] deployed an improved squirrel search algorithm (ISSA) with a modified deep belief network (MDBN), resulting in 99.8% accuracy. Genetic algorithms (GA) were also introduced in intrusion detection systems by Arunkumar et al. [45]. Another study conducted in [46], primarily for feature selection, contributed to improved detection performance. Moreover, the salp swarm algorithm (SSA), another nature-inspired algorithm, was implemented with LSTM [47], achieving an accuracy of 99.71%, while another nature-inspired algorithm, gray wolf optimization (GWO), was implemented with CNN [48], showing competitive performance with an accuracy of 99.12%.

Other studies have introduced novel or hybrid optimization methods. Sreelatha et al. [49] proposed a sandpiper

Table 4 Summary of ML-based cloud IDS studies from a performance optimization perspective (Acc=Accuracy, Prec=Precision, F1=F1 Score, Cat.=Category, Par.=Paradigm, Lear.=Learning)

Cat.	Reference	Algorithm	Dataset	Attacks	IDS	Par.	Lear.	Acc/Prec/F1
Nature Inspired Optimization	[57]	BT, RF	UNSW-NB15	BD, DoS, Expl., Fuz., SC, Recon., Worms	NIDS	SL	ML	99.5/99/99
	[47]	SSAFS, LSTM, AO	Custom	DoS, R2L, Probe, U2R	Hybrid	SL	DL	99.7/94.9/94.5
	[37]	MTL-DNN, CMFO	KDDCup99, NSL-KDD	Generic attacks	NIDS	SL	DL	98.2/69.1/63.0
	[38]	ARSO, BiLSTM	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	SL	DL	98.2/97.5/97.2
	[39]	A-BiLSTM, HHO	UNSW-NB15	BD, DoS, Expl., Fuz., SC, Recon., Worms	NIDS	HL	DL	98.2/69.1/63.0
	[40]	D-BiLSTM, MgOA	UNSW-NB15, BoT-IoT	PS, KL, Exfil., BD, DoS, Expl., Fuz., SC, Recon.	NIDS	HL	DL	99.6/–/98.6
	[41]	FDNN, HBA, PPB	NSL-KDD, CICIDS2017	DoS, Probe, R2L, U2R, BF	NIDS	HL	DL	99.9/99.7/99.7
	[43]	SVM, CSA	KDDCup99	Botnets, DoS, Probe, U2R, R2L	NIDS	SL	ML	87.5/80.0/–
	[44]	ISSA, MDBN	UNSW-NB15	BD, DoS, Expl., Fuzzing, SC, Recon.	NIDS	HL	DL	99.8/–/–
	[48]	CNN, GWO	Not given	Cloud storage attacks	NIDS	SL	DL	99.1/85.9/99.4
	[58]	SAE, HHO	CSE-CIC-IDS2018	DoS, BF, Bot, Infil., SQLi, XSS	NIDS	UL	DL	97.4/90.9/90.8
	[59]	NN, DT, MFA	CSE-CIC-IDS2018	DDoS, Bot, BF, Web, Infil., DoS	NIDS	SL	ML	97.0/96.0/88.9
	[49]	SOA, AlexNet	UNSW-NB15, NSL-KDD	DoS, R2L, U2R, Probe, BD, Expl., Worms	NIDS	SL	DL	99.9/99.9/–
	[50]	SHO, RF	CSE-CIC-IDS2018	DoS, DDoS, Probe, R2L, U2R, BF, Web, Infil.	NIDS	SL	ML	99.1/96.9/96.6
	[51]	WPHW, BWO, KNN	HNAP	Generic attacks	NIDS	SL	ML	80.0/–/–
	[45]	Gannet-OA, SVM	UNSW-NB15	BD, DoS, Expl., SC, Recon.	NIDS	SL	ML	–/98/98
	[46]	GA, GOA, RF	UNSW-NB15, CIC-DoS2019	DoS, Expl., BD, Worms, SQLi, XSS, Bot, WebDDoS	NIDS	SL	ML	99.0/99.0/99.0
	[52]	IPOA, SVM	CICIDS2017, UNSW-NB15	DDoS, Bot, BF, SQLi, Infil., BD, DoS, Expl., Worms	NIDS	SL	ML	99.9/97.4/99.9
	[53]	A-BiLSTM, GOA, DHO	CSE-CIC-IDS2018	DDoS, Bot, BF, Web, Infil., DoS	NIDS	SL	DL	98.8/98.0/97.9
	[60]	S-AdaBoost, SVM	NSL-KDD, CICIDS2017	Generic	NIDS	SL	ML	99.8/100/100
Light weight IDS	[61]	Seq2Seq, BA-BiLSTM, NLP	Logs (Linux, Windows, Hadoop, OpenStack, Apache, CSE-CIC-IDS2018)	DDoS, DoS, Bot, BF, Infil., Web, Logs anomalies	Hybrid	SL	DL	99.6/–/–
	[62]	D-CNN	LIDDS-2019	Heartbleed, SQLi, DoS, MITM	HIDS	SL	DL	98.12/98.18/95
	[63]	LR, L-SVM, MLP, RF	ADFA-LD	Linux exploits, Malware, Privilege escalation	HIDS	SL	ML	–
	[64]	NCGTM, GBM, RF, XGBoost, DNN	NSL-KDD, Kyoto2006+	DoS, DDoS, Probe, U2R, R2L, Anomalies	NIDS	SL	ML/DL	82.5/–/–
	[65]	DMA, NN, DT, KNN, Bagging	NSL-KDD, Kyoto2006+	DoS, DDoS, Probe, U2R, R2L, Anomalies	NIDS	SL	ML	99.75/–/–
	[66]	GraphSAGE, CBLOF, IForest	Custom (sim)	Resource/computing OU, MB	HIDS	UL	ML	99.0/79.2/86.4

Table 4 (continued)

Cat.	Reference	Algorithm	Dataset	Attacks	IDS	Par.	Lear.	Acc/Prec/F1
	[67]	DDQN, PER, AdaBoost, RF, KNN, QDA, GNB	ISOT-CID, NSL-KDD	IP Spoofing, DDoS, CM, Probe, U2R, R2L, DoS, Adv.	NIDS	HL	ML/DL	96.8/–/–
	[68]	F-FS, SVM	NSL-KDD, Kyoto2006+, CSE-CIC-IDS2018	DoS, Probe, U2R, R2L	NIDS	SL	ML	99.8/93.02/92.9
	[69]	LS2DNN, C2MJOA-FS, PBKA, FL	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	FL	DL	97.2/96.9/97.0

optimization (SPO) strategy with Extended equilibrium deep transfer learning (EEDTL), achieving 99.9% accuracy. Mary et al. [50] combined sea horse optimization (SHO) with a deep echo state network-based IDS (DESNID), yielding 99.14%. A whale–Pearson hybrid wrapper with KNN was presented in [51], achieving 80% accuracy. In addition, hybrid ML optimization frameworks such as those in [52–54] achieved consistently high accuracies ranging from 98% to 99.86%, demonstrating the growing effectiveness of combined heuristic and learning-based approaches in designing robust IDS for cloud computing.

Lightweight IDS: In the context of cloud computing, a lightweight IDS refers to computationally inexpensive systems (requiring less memory, power, or CPU utilization) and suitable for resource-constrained cloud environments, such as virtual machines (VMs) or containerized environments [55]. The security mechanisms that consume excessive resources are impractical and unsuitable for real-world deployment [56]. Table 4 shows the optimized IDS approaches designed to operate efficiently.

The study [61] proposed a cloud IDS, a Seq2Seq model based on BiLSTM and Bahdanau attention, achieving 98.2% accuracy on a multi-domain dataset and 94.2% with CSE-CIC-IDS2018 and LogHub datasets [70]. A lightweight host-based IDS was introduced for containerized environments using CNN on image-based logs, and Apache Spark memory-based cluster computing engines in [62] and [63], respectively.

A game theory-based lightweight IDS was proposed in [64], modeling attacker-defender interaction and showing an improved detection rate and lower false positives via a stacked ML framework. Another approach in [65] applied the dolphin mating algorithm (DMA) for feature selection, reducing computational time and improving detection accuracy. Moreover, a study [66] proposed a knowledge graph-embedding-based anomaly detection system focused on optimizing resource allocation in cloud environments.

The authors in [67] used a double-DQN (DDQN) to reduce the computational capacity of the proposed IDS, achieving 96.87% accuracy on ISOT-CID and NSL-KDD. A filter-based SVM approach was presented in [68], which demonstrated improved efficiency across NSL-KDD, Kyoto, and CSE-CIC-IDS2018 datasets.

3.1.2 Learning and adaptation perspective

The learning and adaptation perspective includes primary studies that enhance an IDS's ability to respond to evolving cloud behaviors and threat patterns. This perspective covers FL–based cloud IDS, anomaly-driven detection, adversarially resilient methods, and traditional ML approaches. The following subsection discusses primary studies that fall under this perspective.

Federated learning-based IDS: In the cloud computing context, FL is a distributed ML-based approach that enables the distributed training of models across multiple cloud nodes or edge devices without sharing private data [111]. IDS approaches that integrate FL to enhance privacy and security in cloud environments are listed in Table 5.

The work in [71] proposed a decentralized IDS using federated averaging (FedAvg) and a CNN, trained on CICIDS2017 with client-specific IP mapping. The system achieved 93% accuracy while preserving user privacy. Similarly, the authors in [69] introduced a privacy-preserving FL-based IDS using the LS2DNN classifier and Pearson-Brownian K-anonymity (PBKA), validated on NSL-KDD and outperformed baseline models with 97.24% accuracy. The authors in [72, 73] extended FL to multi-tenant cloud environments using differential privacy (DP)-FL, secure multi-party computation (SMC)-FL, and homomorphic encryption (HE)-FL to prevent data leaks and interface-level attacks.

Anomaly detection-based IDS: An anomaly-based IDS refers to an IDS that detects outliers or unusual patterns deviating from normal (expected) behavior [112]. Various studies have emphasized anomaly detection-based IDS in cloud platforms, as summarized in Table 5.

The methodology in [74] combined an autoregressive chimp optimization algorithm (ACHOA) with a deep Q-network (DQN) to identify unusual patterns. Similarly, the approach in [75] proposed an adaptive anomaly detection system for IaaS-level VMs using Zoom-CNN, DQN,

Table 5 Summary of studies on learning and adaptive perspective (Adv.=Adversarial, Trad.= Traditional)

Cat.	Reference	Algorithm	Dataset	Attacks	IDS	Par.	Lear.	Acc/Prec/F1
Federated Learning-based IDS	[69]	LS2DNN, C2MJOA-FS, PBKA	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	FL	DL	97.2/96.9/97.0
	[71]	FedAvg, CNN	CICIDS2017	BF-FTP, BF-SSH,(D)DoS,GE, Hulk,SHT,HeartBleed, Web,Infil.,Bot	NIDS	FL	DL	93/–/–
	[72]	FedAvg, LLMs, SCL, HE	Google Cloud BigQuery	Generic	NG	FL	ML/DL	85/–/–
	[73]	FedAvg, FedProx, SF-XGBoost, DP, SA	Credit Default	Generic	NG	FL	ML	97.63/1.00/0.84
Anomaly Detection-based IDS	[74]	DQN, RL, ARCO	BoT-IoT, KDDCup99	Network Anomalies	HD	RL	RL	94.0/–/–
	[75]	DQN, Z-CNN	Real IaaS VM Monitor	Memory Anomalies, Resource Anomalies, VM Anomalies	HIDS	HL	DL	99.8/97.4/98.13
	[76]	XGBoost, LightGBM, AdaBoost, RF, BMA	KDDCup99, UNSW-NB15, CICIDS2017, NSL-KDD, TRAbID	General Network Attacks, Network Anomalies	NIDS	SL	ML	96.8/97.0/97.0
	[77]	LR, KNN, DT, ET, RF, GB, MLP	UNSW-NB15	BD, DoS, Expl., Fuz., SC, Recon., Worms, Network Anomalies	NIDS	SL	ML	97.6/97.6/96.3
	[78]	CT, EBT, Q-SVM, BT	UNSW-NB15	BD, DoS, Expl., Fuz., SC, Recon., Worms, Network Anomalies	NIDS	SL	ML	81.0/92.0/92.0
	[79]	SGD, LR, Ridge, RF, XGBoost, GB, CNN-LSTM	UNSW-NB15	BD, DoS, Expl., Fuz., SC, Recon., Worms, Network Anomalies	NIDS	SL	ML/DL	99.91/98.3/98.4
	[80]	FS + Prophet Model	CSE-CIC-IDS2018	DoS, DDoS, BF, PS, Network Anomalies	NIDS	SL	ML	–
	[81]	ELBD, IForest, DT, KNN, LOF, OCSVM, MLP	Server Machine, Vichalana	Anomaly Detection	HIDS	HL	ML/DL	–/–/83.2
	[82]	XGBoost, RNN	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	SL	ML/DL	99/99/99
	[83]	DQN, DRL, RF, AdaBoost, KNN, GNB, QDA	UNSW-NB15	DoS, BD, Expl., Fuz., Recon., SC, Worms, GE, Analysis	NIDS	HL	RL/ML	83.80/–/–
	[84]	RF, Feature Engineering	NSL-KDD, BoT-IoT	Anomaly Detection	NIDS	SL	ML	98.3/96.3/–
Adv. Resilient IDS	[85]	SHAP-FS, RF, XGBoost	CSE-CIC-IDS2018	FGSM, PGD, DeepFool, Square Attack	NIDS	SL	ML	88/–/–
	[86]	DD, AT, GAI, XAI	No dataset	Adversarial Attacks	NIDS	SL	DL	NG
	[87]	BiGAN, DNN	NSL-KDD	Adversarial, Evasion	NIDS	HL	DL	NG
	[88]	DD, AT	KDDCup99, CICIDS2017, DARPA	Evasion	NIDS	SL	ML	98/–/–
Trad. ML-Based IDS	[24]	CDAAE, SVM, CDAAE-KNN, DT, RF	Cloud DoS, NSL-KDD, UNSW-NB15, CICIDS2017	HTTP-based DDoS, Low-rate DDoS	NIDS	HL	ML/DL	NG
	[89]	Conv1D, GRU	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	SL	DL	96.57
	[90]	OBCLSTM, ERFO	NSL-KDD, TUIDS, UNSW-NB15, BoT-IoT	DoS, Probe, U2R, R2L, Malware	HD	HL	ML/DL	96.57/86.2/98.3
	[91]	DAE, CNN	CIC-DDoS2019	Low-rate DDoS	NIDS	HL	ML	DR=95.31
	[92]	KFCM, T2FNN, LO	NSL-KDD	IP Spoof, RIP, Flooding, DoS, DDoS	NIDS	SL	ML	98.45/–/96.0
	[93]	RF, SMOTE, IG, CS, PSO	Kyoto2006+, UNSW-NB15	DDoS, Probe, U2R, R2L, BD, Expl., Fuz., SC, Recon., Worms	NIDS	SL	ML	99.2/99.2/99.1
	[94]	B/BT, SD, RUSBoost, VS	CICIDS2017	DDoS, PS, BF, Infil.	NIDS	SL	ML	97.4/97.0/–

Table 5 (continued)

Cat.	Reference	Algorithm	Dataset	Attacks	IDS	Par.	Lear.	Acc/Prec/F1
	[95]	KMC, KNN, DT, SVM	IDSStore	Misconfiguration Attacks	NG	HL	ML	82/70/75
	[96]	DT, NB, MV	NSL-KDD	DoS, Probe, U2R, R2L	NIDS	SL	ML	99.7/99.8/99.8
	[97]	SVM, KNN, H-CNN–LSTM	CICIDS2017, NSL-KDD	DoS, Probe, U2R, R2L, BF, HB, Web, Infil.	NIDS	SL	ML/DL	99.0
	[98]	GAN, LightGBM, ADASYN	UNSW-NB15, NSL-KDD	DoS, Probe, U2R, R2L, BD, Expl., Fuz., SC, Recon., Worms	NIDS	HL	ML/DL	93.3/98.3/92.2
	[99]	DNN, RF, KNN, NB, LR, MLP	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	SL	ML/DL	99.98/99.3/–
	[100]	SVM, DT	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	SL	ML	NG
	[101]	TB model	Custom dataset	HTTP-based DDoS	NIDS	HL	ML	100
	[102]	AdaBoost, RF, DT, SGD, TFL, TFBT, D-ANN	Custom dataset	SQLi	HIDS	SL	ML/DL	99.8/99.9/99.9
	[103]	AdaCyborg, RF, BN, C5.0, CART	KDDCup99	Botnets, DoS, Probe, U2R, R2L	NIDS	SL	ML	99.8/99.5/99.2
	[104]	FRC, EBSA	Not mentioned	Generic attacks	NIDS	SL	ML	99.85/–/99.76
	[105]	FLM, KNN, KSM	Custom session vectors	Application-layer Attacks	NIDS	SL	ML	96.25/95.5/93.5
	[106]	UEFFS, SVM, NB, LR, DT, MV	Honeypot, Kyoto2006+, NSL-KDD	DoS, Probe, U2R, R2L	NIDS	SL	ML	99.9/99.9/99.9
	[107]	HT-ML, SFE, HE, ECC, HMM	Real-world datasets	DDoS, BF, Infil., Malware, Phishing	NIDS	SL	ML/DL	98.6/96.8/98.0
	[57]	BT, RF	UNSW-NB15	BD, DoS, Expl., Fuz., SC, Recon.	NIDS	SL	ML	99.5/99.0/99.0
	[108]	LR, RF, DT	CSAD (Kaggle)	Application-layer Attacks	HD	SL	ML	99.8/99.5/99.2
	[47]	SSAFS, LSTM, AO	Custom dataset	DoS, R2L, Probe, U2R	HD	SL	DL	99.7/94.9/94.5
	[109]	MN-v2, SSD, SN	WIDER Face, RMFD	Identity-based	HIDS	SL	DL	98.8/95.4/–
	[110]	LSTM, SCAE, SVM	NSL-KDD	DoS, Probe, R2L, U2R	NIDS	SL	DL/ML	99.6/100/100

and transfer learning. Boosting techniques were explored in [76], including AdaBoost, LightGBM, and XGBoost, to improve the detection performance using the UNSW-NB15 dataset.

In addition, a secure packet classifier (SPC) using collaborative filtering with two ML algorithms, achieving 94% accuracy and a 92% F1 score, was presented in [78]. CNN-LSTM-based anomaly detection was explored in [79], achieving 97.06% (binary) and 99.91% (multi-class) accuracy. The researchers in [80] employ the Facebook Prophet time-series model for predictive anomaly detection in cloud traffic. The ensemble and hybrid models also showed effectiveness ([76, 77, 81, 82]), yielding an ARP score of 5.1821 and achieving 99% accuracy.

Adversarial-resilient IDS: Adversarial-resilient IDS refers to intrusion detection systems capable of resisting adversarial attacks (attacks on ML models) while maintaining a reasonable detection accuracy [113]. Table 5 presents a summary of the relevant primary studies that discussed IDS approaches developed to protect against adversarial attacks in cloud environments.

The researchers in [85] proposed a dual-defense strategy combining adversarial training and SHAP to protect against both white-box attacks and black-box square attacks. Validated on the CSE-CIC-IDS2018 dataset, the framework achieved 88% detection accuracy for gradient-based and 84% for black-box attacks, reducing the adversarial success rate by up to 40%.

Another approach proposed in [114] introduced a novel attack, a fast feature loss (FFL)-PGD attack using ResNet-152 to generate adversarial examples targeting cloud APIs, achieving over 90% success with minimal queries. The researchers suggested two types of training, online and offline, that used tools like AdvBox to improve the resilience of their model and limit how easily an attacker could move its "adversarial noise" around. Another study [86] examined ways to defend against adversarial attacks by developing generative artificial intelligence (AI) in the cloud, providing a multi-layered defense system consisting of adversarial training, defensive distillation, encryption, and continued anomaly detection. Some studies have also developed detection-based defenses. For example, a study published in [87] developed a bidirectional GAN-based

adversarial sample detector (ASD) that improved adversarial detection accuracy from 55% to 82% on the NSL-KDD dataset. Also, a study published in [88] tackled evasion attacks using ensemble classifiers trained on reduced feature sets (RFS) created via permutation feature importance (PFI), demonstrating high accuracy across the tested datasets, including KDDCup99, CICIDS2017, and DARPA-Hydra.

Traditional ML-based IDS: This subsection reviews IDS studies that apply traditional ML and DL techniques to detect a broad range of attacks in cloud environments using publicly available datasets. Please note that we created this as a separate category to discuss the studies where ML models were used without any additional methods (as discussed in the performance optimization perspective) to optimize performance, reduce training time, or improve efficiency. These studies focused solely on identifying the model that performed best for detecting specific attacks. A summary of these traditional ML-based approaches is presented in Table 5.

In the study [89], a hybrid CNN-RNN model was used on the NSL-KDD dataset. In contrast, [90] introduced a Bi-CNN-LSTM model (OBCLSTM) evaluated on NSL-KDD, TUIDS, UNSW-NB15, and BoT-IoT, achieving a 99.1% detection rate. Similarly, the authors in [91] used deep autoencoders with CNN to detect low-rate DDoS attacks, attaining 95.32% accuracy. Fuzzy and clustering-based techniques were explored in [92] using kernel fuzzy C-means with a type-2 fuzzy neural network, achieving 98.45% precision and recall. The study in [54] combined FCM with SVM for U2R and probe-type attack detection on NSL-KDD, achieving an accuracy of over 97%. The researchers in [93] used RF on the UNSW-NB15 and Kyoto datasets, achieving a 99% multi-class accuracy. Moreover, another approach presented in [97] explored combinations of KNN and CNN-LSTM and was validated on CICIDS2017 and NSL-KDD datasets.

The study in [98] addressed class imbalance using a generative adversarial network (GAN) and lightweight gradient boosting (Light-GBM), achieving binary classification with high accuracy, though risking overfitting. Many studies have proposed ensemble-based methods, as they contribute to the effectiveness of an IDS. For instance, the study in [94] evaluated testing boosted and bagged classifiers on CICIDS2017, whereas the study in [96] combined DT with Naive Bayes, achieving 90% on NSL-KDD. Moreover, the study discussed in [95] proposed the concept of IDSStore, which uses a hybrid K-means and DT approach for an F1-score of 82%. All of these studies used an ensemble-based method for IDS.

Similarly, the study presented in [24] used deep generative models, a conditional denoising adversarial autoencoder (CDAAE), and its hybrid with KNN (CDAAE-KNN) to synthesize malicious samples. Another approach discussed in [102] is the detection of SQL injection using ensemble ML classifiers. The researchers in [106] combined a univariate ensemble filter with ensemble learners using Honeypot, Kyoto, and NSL-KDD datasets, achieving 96% accuracy. The study in [99] compared the results of multiple ML models, finding the best combination of DNN and RF with 99.7% accuracy. The approach used in [104] introduced an effective IDS and classification using a fuzzy rule-based classifier (EIDC-FRC) model in a cloud environment, achieving 99.76% F-measure. The study conducted in [105] applied nearest-neighbor and keyword-based ML, with 96.25% accuracy. However, [107] introduced cryptographic hyperparameter tuning with an accuracy of up to 98.9%.

The study in [110] applied LSTM, SCAE, and SVM, with LSTM exhibiting better performance. The authors in [83] developed a deep reinforcement learning (DRL)-based IDS, achieving 83.07% accuracy on UNSW-NB15. Another study conducted in [84] proposed a cloud-specific RF-based IDS with visualization, reaching 98.3–99.9% accuracy across BoT-IoT and NSL-KDD.

3.1.3 Environment-aware perspective

The environment-aware perspective includes primary studies that design IDS solutions to operate effectively within specific cloud architectural layers. This perspective evaluates studies on multi-cloud IDS, which address detection across diverse and distributed cloud platforms, and hypervisor-based IDS, which focus on detecting attacks within virtualized infrastructures. The following subsection discusses primary studies that fall under this perspective.

Multi-cloud IDS: A multi-cloud setup uses services from several cloud providers to improve flexibility and reliability, whereas a hybrid cloud connects private and public clouds to balance scalability, cost, and data control [123]. This subsection reviews ML-based IDS studies specifically designed for multi-cloud environments. Table 6 summarizes the relevant studies.

The study in [115] demonstrated that the use of federated identity management, homomorphic encryption, and risk-based access controls for an IDS increased its performance by 15–20% when compared to other traditional IDSs. It also utilized real-world multi-cloud incident data sets as validation for this approach. Work completed in [116] created a hybrid DRL framework using a denoising autoencoder and Q-learning. This architecture achieved 99.1% detection accuracy with a 98% F1 score on three different datasets: NSL-KDD, KDDCup99, and Yahoo A1/A2. In [117], the researchers developed a hybrid multi-cloud setting with an AI-driven IDS. They used a combination of CNN, RNN,

Table 6 Summary of literature on environment-aware perspective (Acc=Accuracy, Prec=Precision, F1=F1 Score, Cat.=Category, Par.=Paradigm, Lear.=Learning, HV=Hypervisor)

Cat.	Reference	Algorithm	Dataset	Attacks	IDS	Par.	Lear.	Acc/Prec/F1
Multi-cloud IDS	[115]	HE, FIM, IDS, RBAC	AWS, Azure, GCP, IBM Cloud Logs	DDoS, SQLi	NIDS	SL	ML	NG
	[116]	DRL, DQN	NSL-KDD, KDDCup99, Yahoo A1/A2	DoS, Probe, R2L, U2R	NIDS	SL	DL	95/–/–
	[117]	CNN, RNN, RL	Simulated Multi-cloud Logs	DDoS	HIDS	SL	ML/DL	NG
	[118]	I-GN, OBL-RIO, 2D-ACNN	NF-UQ-NIDS (NF-UNSWNB15, NF-BoTIoT, NF-ToNIoT, NF-CSE-CIC-IDS2018)	DDoS, DoS, Recon., BD, IL, MITM, Ransomware, XSS, PS, Exploits, Fuzzers, Shellcode, Worms, Bot, BF, Web, SSH Unauth., GE, Hulk, SHT, SL, Infil., SQLi	NIDS	SL	DL	95.20/–/–
HV-based IDS	[54]	FCM, SVM	NSL-KDD	DoS, U2R, Probe, R2L	Hybrid	SL	ML	97.3/96.2/97.0
	[119]	FL	CICIoT2023, UNSW-NB15	VM Escape, Side-Channel, Hyperjacking	HIDS	FL	FL	92.6/92.1/92.0
	[120]	ERBNN, AFF	KDDCup99, NSL-KDD	Generic	Hybrid	SL	DL	97.1/97.2/97.1
	[121]	EM, FTS, AFF	Synthetic, KDDCup99	DDoS Attacks	NIDS	UL	ML	98.46/–/–
	[122]	RLDAC-IDS	Custom dataset	Zero-day, VM Escape, DDoS	Hybrid	RL	RL	98.7/98.5/97.5

RL, and a mathematical optimization model. Their IDS successfully identified DDoS, insider threats, and data breach attacks. In [118], Nizamudeen et al. described a more specific IDS. This IDS used integrated normalization (IGN) and opposition-based learning rat-inspired optimization (OBL-RIO) for feature selection. Following feature selection using OBL-RIO and IGN, they used a two-dimensional CNN classification model to identify anomalous traffic patterns. Their experimental results showed that the classifier had excellent performance (accuracy of 95.2%, detection rate of 97.24%) and also had an extremely low false positive rate (only 2.5%) on the NF-UQ-NIDS dataset. This work represents a new direction in research and the necessity for developing flexible IDS architecture models that can be adapted to increasingly distributed and complex cloud computing environments.

Hypervisor-based IDS: A hypervisor is a virtualization management layer that allows multiple VMs to share the same physical environment while ensuring isolation, efficient resource allocation, and secure execution in cloud environments [124]. This section presents an analysis of studies that proposed an IDS to detect attacks on hypervisors (shown in Table 6).

A hypervisor inspector was developed using a combination of SVM and FCM clustering, achieving high detection accuracy for U2R (97.37%), R2L (98.46%), and probe attacks (98.85%) on the NSL-KDD dataset in [54]. The study in [119] proposed an FL-based IDS to detect hypervisor-specific threats such as hyperjacking, side-channel attacks, and VM escapes. Their hybrid model achieved 92.6% accuracy on CICIoT2023 and UNSW-NB15.

An enhanced radial basis neural network (ERBNN) optimized with the adaptive artificial flora (AAF) algorithm was presented in [120], delivering up to 99.87% accuracy. Another study proposed in [122] introduced an RL-based self-adaptive IDS that integrates both anomaly- and signature-based methods at the hypervisor level, reaching 98.7% accuracy. Finally, another study in [125] introduced "Guillotine", a hypervisor-level architecture featuring multi-layered sandboxing-level kill switches, ranging from software to hardware isolation and datacenter. A real-time black-box detector dynamically adjusts enforcement levels to neutralize threats and proactively enhance AI system safety.

Overall, the proposed taxonomy provides several practical insights for researchers and practitioners working on cloud IDS. First, the large concentration of studies under the performance optimization perspective, particularly nature-inspired optimization (20 studies) and traditional ML-based IDS (25 studies), indicates that improving detection efficiency and scalability remain dominant research priorities. Nevertheless, the consistent use of legacy dataset sources, including NSL-KDD (26 studies) and others, within these research categories significantly reduces the real-world validity of demonstrated performance improvements and should be addressed with prioritized evaluations of current, cloud native benchmark datasets.

Additionally, as researchers are increasingly developing learning and adaptive IDS approaches (including anomaly-based and FL), there is an increasing demand for IDSs that can learn from new and evolving attacks and also operate across distributed cloud environments. Notably, while approximately 74.7% of all studies employ supervised learning methods, only 3.8% employ unsupervised methods. This represents a research imbalance towards labeled attack scenarios. As a result, the potential of many current

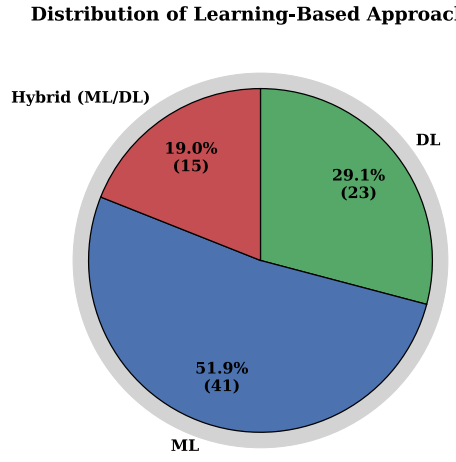


Fig. 3 Distribution of learning-based approaches adopted in cloud-based IDS

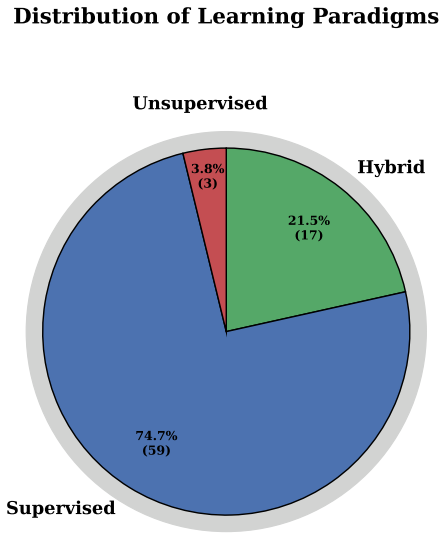


Fig. 4 Distribution of learning paradigms applied in ML-based IDS for cloud environments

IDS approaches to provide effective protection against zero-day and emerging threats will likely be limited.

Further, while cloud-based IDS approaches such as adversarial-resilient IDS, hypervisor-based IDS, and multi-cloud IDS have emerged as important over recent years, they remain comparatively underexplored, represented by only 4, 5, and 4 studies, respectively. Therefore, researchers will need to concentrate their next efforts on the development of stronger detection methods that are resilient against adversary attacks; increasing visibility and tracking functionality in the virtualization layer; and designing interoperability mechanisms that will facilitate communication among different, heterogeneous cloud environments. As such, this classification is intended to provide an outline for identifying both established research avenues and additional avenues for cloud-specific IDS research.

3.2 Findings based on primary studies

This section presents a comprehensive review of the research (presented in the preceding section) that focuses on the current ML landscape for cloud IDS; the types of attacks studied; and the datasets utilized to assess the applicability of existing cloud IDS approaches. Performance metrics shown in Tables 4, 5 and 6, which were taken directly from the corresponding source study, are representative of the performance of various experiments utilizing different datasets, attacks, techniques for feature extraction, preprocessing techniques, and test protocols. Therefore, it is essential to view these performance metrics with respect to their individual experiment context, rather than comparing them in terms of specific benchmarks.

3.2.1 Classification of primary studies based on ML landscape

We evaluated 79 primary studies based on the scope defined for this SLR (Sect. 2). This subsection discusses the ML landscape in the context of the ML algorithm used, ML paradigms, and adopted learning approaches.

Distribution of learning-based approaches: Fig. 3 shows the statistical analysis of learning-based (ML-based, DL-based, and hybrid) approaches utilized in cloud-based IDS research.

The majority of the studies, almost 52%, have used machine learning techniques, whereas 29% have applied DL-based approaches, and the remaining 19% have utilized hybrid ML/DL models for designing the IDS. The most frequent use of ML approaches highlights their maturity level and suitability for intrusion detection scenarios.

Distribution of ML paradigms: The next analysis is the examination of the studied literature with respect to learning paradigms, which will help in a better analysis of applications of ML in a cloud environment. Figure 4 shows the statistical analysis of learning paradigms, where 74.7% (59 studies) are based on supervised learning.

Moreover, 21.5% (17 studies) are based on hybrid approaches, and only 3.8% (3 studies) utilize unsupervised learning. The frequent use of supervised learning is largely due to the wide availability of labeled datasets. This limited application of unsupervised learning demonstrates an obvious study focus on labeled data. This distribution shows that methodological support for detecting zero-day threats and previously unseen threat types is extremely low

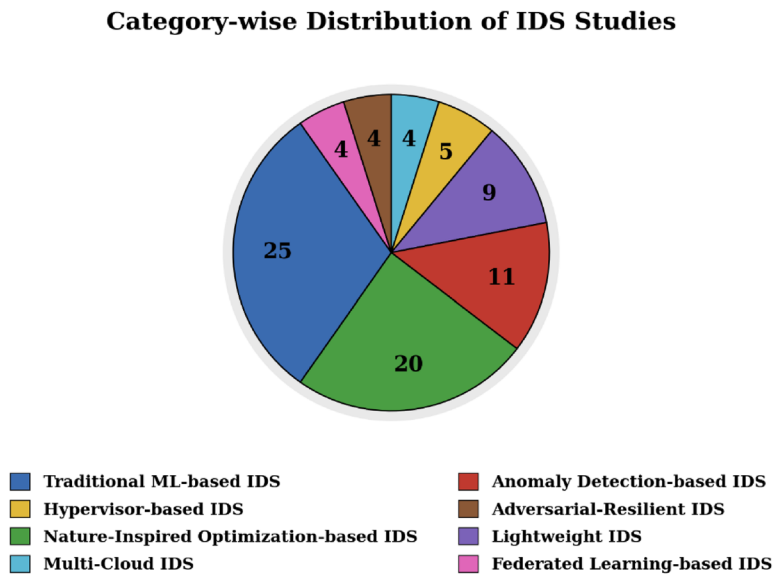


Fig. 5 Distribution of studies with respect to categories

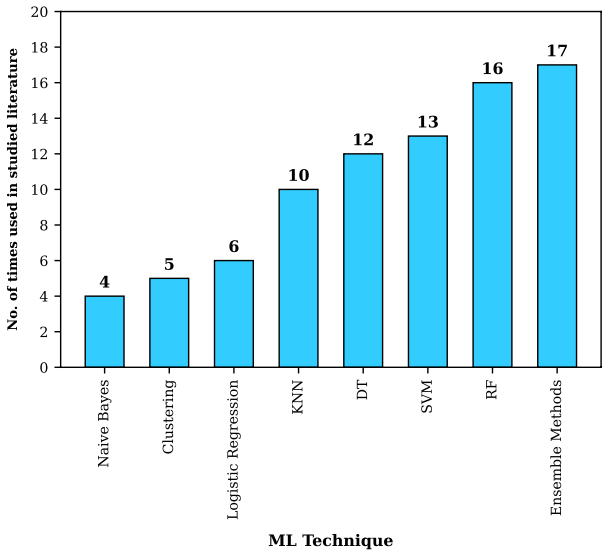


Fig. 6 Classification of machine learning approaches applied in cloud-based IDS

in the studied literature of cloud-based IDSs, which typically require approaches that can detect new behaviors and patterns outside of what was defined by supervised labeled training sets.

Category-wise distribution of ML-based IDS studies: Fig. 5 presents the category-wise classification of ML-based IDS studies according to their methodological orientation and design focus. The largest portion of research falls under the traditional ML-based IDS and nature-inspired optimization-based IDS, comprising 25 and 20 studies, respectively. These categories dominate because of their proven efficiency and adaptability to network traffic patterns. A moderate number of studies address anomaly detection-based and lightweight IDS designs, emphasizing efficient detection in constrained environments. In contrast, only a few studies have explored hypervisor-based, multi-cloud, adversarial-resilient, and FL-based IDS frameworks, reflecting emerging yet underdeveloped research directions in cloud security.

Classification of ML techniques in cloud-based IDS: Fig. 6 summarizes the ML approaches adopted in cloud-based IDS research.

The results show that models such as SVM, Random Forest, Decision Trees, KNN, Logistic Regression, Naive Bayes, and ensemble methods are mainly used for classification in cloud-based IDS, highlighting their effectiveness in distinguishing between normal and malicious traffic. Moreover, a moderate number of studies have applied clustering techniques for anomaly detection. Several studies have explored ensemble methods to enhance adaptability and detection performance; however, the overall trend favors supervised classification, whereas advanced learning approaches remain limited in cloud-based IDS.

Classification of DL techniques in cloud-based IDS: Fig. 7 illustrates the distribution of DL architectures adopted in cloud-based IDS studies.

The analysis revealed a preference for sequence and pattern learning models within a cloud-based IDS. CNN appears most frequently (11 studies), followed by LSTM (9), DNN (6), and RL (6), reflecting their ability to capture complex temporal, spatial, and adaptive patterns in attack traffic. Meanwhile, only a limited number of studies have explored the use of autoencoders, GANs, and GraphSAGE. Overall, the distribution highlights a dominant reliance on CNN- and LSTM-based models, whereas more complex or hybrid DL frameworks have been less explored in cloud IDS research.

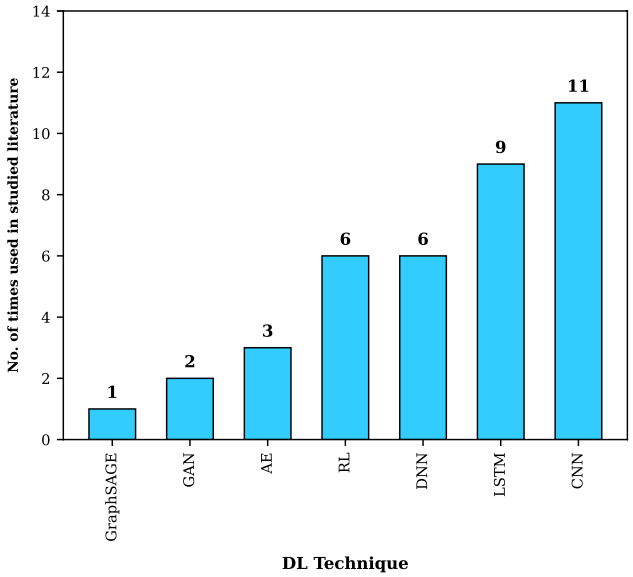


Fig. 7 Classification of deep learning approaches for IDS applied in the cloud

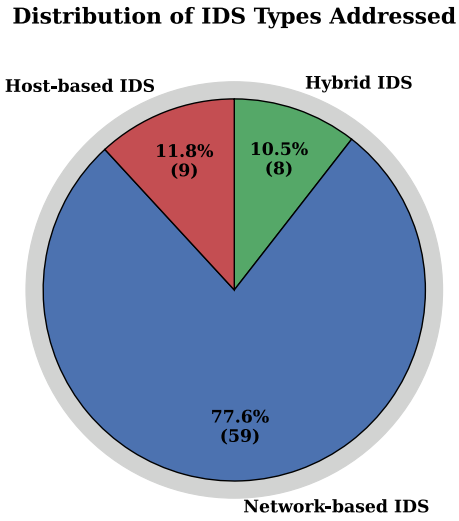


Fig. 9 Distribution of studies with respect to IDS types

3.2.2 Classification of primary studies based on security attacks detected, IDS types, and utilized datasets

Distribution of security attacks detected by IDS: Fig. 8 presents the classification of the reviewed studies according to the types of security attacks detected within the cloud environment.

The analysis shows that DoS and DDoS attacks dominate the literature and represent the most frequently investigated threat category. A moderate number of studies focus on reconnaissance, R2L, U2R, backdoor, brute-force, and other exploit-based attacks, while comparatively fewer studies have addressed SQL injection, XSS, SSH, adversarial evasion, and hyperjacking attacks. This indicates that most IDS models remain oriented toward traditional network and system attacks, whereas emerging threats in virtualized cloud infrastructures remain comparatively less explored.

Distribution of studies with respect to IDS types: Fig. 9 shows a clear pattern in how IDSs are being used in recent studies.

Most of the work still focuses on network-based IDS, with 59 studies (about 77.6%) targeting traffic monitoring and network-level attacks in cloud environments. In comparison, host-based IDS appeared in 9 studies (approximately 11.8%), and hybrid IDS approaches were mentioned in 8 studies (approximately 10.5%). Overall, the statistics highlight that current research continues to lean heavily toward network-centric detection, whereas host-level and combined approaches receive far less attention.

Classification of studies based on utilized datasets:

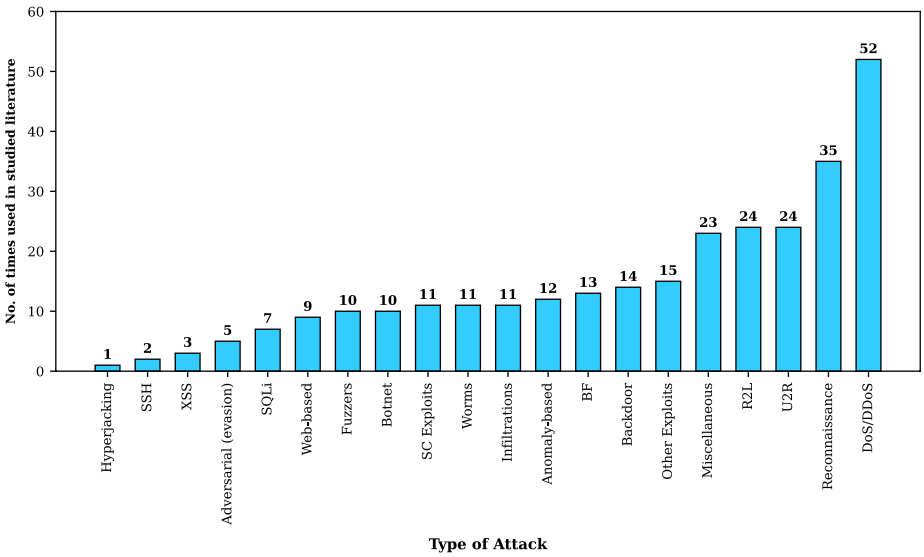


Fig. 8 Distribution of security attacks detected by IDS in cloud environments

Figure 10 illustrates the distribution of the various datasets that have been utilized by the researchers who trained and tested the ML-based IDS models reported in this work. As indicated in the figure, datasets are significant to the research community and especially important to supervised learning-based approaches to IDS modeling. Model performance is heavily influenced by both the quality of the data and the degree of variety present in the labeled data. The numbers indicate dataset-level mentions extracted from Tables 4, 5 and 6; therefore, a single study could potentially reference more than one dataset bar if it utilizes more than one dataset. Apart from frequently used datasets, we categorized other reported datasets into three categories: Other Public Benchmarks, custom/simulated, and cloud/Platform Logs. Those studies that did not make explicit mention of a dataset were excluded from this frequency calculation of datasets. The data show that many of the studies in this work used either infrequently used or non-cloud-specific datasets, e.g., other public benchmarks, custom/simulated local traffic, and cloud-platform logs, indicating a lack of readily available standard benchmarking datasets for use with cloud environments. Furthermore, we found that 26 studies are using the NSL-KDD dataset, which is an older dataset and lacks a comprehensive depiction of contemporary cloud-specific attack patterns.

We observe that the reported performance of IDSs strongly depends on the datasets that are used to evaluate them. Studies utilizing legacy or curated benchmark datasets, e.g., NSL-KDD, KDDCup99, UNSW-NB15, and CICIDS2017, typically report extremely high and often nearly ceiling accuracy values. However, since these types of datasets are typically curated, extensively reused, and include relatively static feature spaces and attack distributions, the results of such studies need to be viewed critically. On the other hand, studies that utilize cloud/platform logs, container logs, customized or simulated data, or multi-cloud traffic typically encounter much greater heterogeneity in traffic patterns, noisier features, less uniform labeling, and more varied or poorly documented performance metrics.

The same observation can be made from Fig. 10, where cloud/platform logs only appeared five times whereas legacy/public benchmark datasets were referenced more frequently. An example of this can be seen in the study conducted by Kim et al. [61], where they applied an identical lightweight cloud IDS in two separate evaluations. The reported accuracy was 98.2% using a multi-domain benchmark dataset, but dropped to 94.2% using a mixed dataset containing CSE-CIC-IDS2018 and LogHub logs. Since the same model and pipeline were employed in both cases, comparing the two evaluations removes some sources of variability that could otherwise affect comparisons among studies, indicating that the differences in reported accuracy are likely due to differences in dataset characteristics. Thus, high accuracy on legacy or curated benchmark datasets should primarily be taken as indicative of within-dataset effectiveness rather than as evidence of generalizability to modern cloud environments. This observation provides additional support for the dataset dependency limitation mentioned in Observation 5.

Many of the primary studies reviewed in this paper rely on benchmark datasets like NSL-KDD, KDDCup99, UNSW-NB15, and CICIDS2017/18, which were created prior to the release of the MITRE ATT&CK Cloud Matrix. Therefore, none of these legacy benchmark datasets natively include ATT&CK technique labels. Consequently, the mapping described in this paper was established through identifying

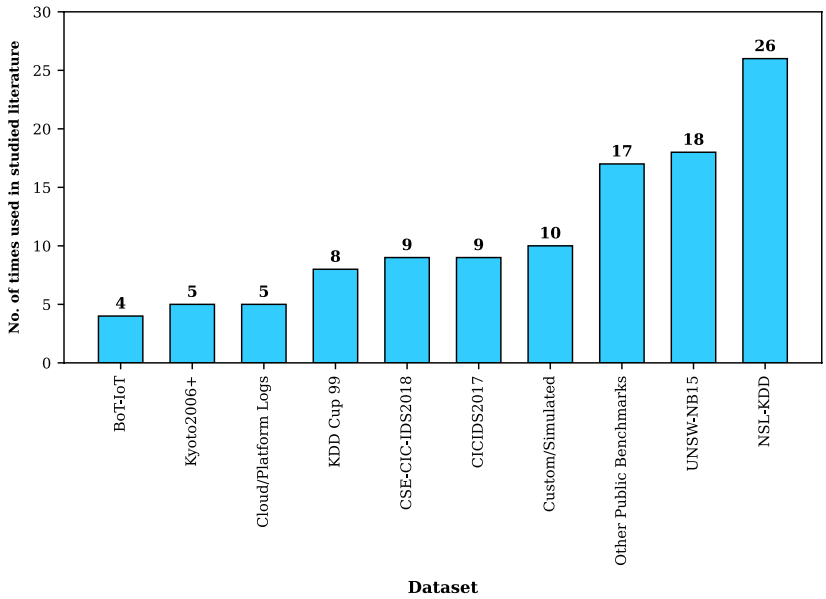


Fig. 10 Statistics of datasets used in primary studies

the behavioral intent and operational goals associated with each attack type category, rather than relying on the original label for each dataset. More specifically, attack types were examined based upon their underlying adversarial behavior and then matched up with corresponding tactics and techniques in MITRE ATT&CK Cloud. For instance, denial-of-service attacks were assigned to the Impact tactic based on their fundamental goal of disrupting services. Denial-of-Service behaviors at the host level were mapped to Endpoint Denial of Service (T1499), whereas denial-of-service flooding behaviors occurring over networks were mapped to Network Denial of Service (T1498). When applicable cloud-specific techniques were not identified during this process, the most analogous behaviorally-related MITRE ATT&CK Cloud technique was chosen. By doing so, this analysis enables an accurate interpretation of legacy-benchmark attacks within a contemporary cloud-threat security model.

4 Existing ML-based cloud IDS and cloud security requirements

This section discusses the following research question. **RQ2.***To what extent do the existing ML-based IDS approaches meet real-world cloud security requirements?*

4.1 Alignment of identified attacks with the MITRE ATT&CK cloud matrix

4.1.1 Mapping motivation

As cloud computing environments grow in scale and complexity, the associated threat landscape continues to evolve, with attackers increasingly employing stealthy and multi-stage tactics. To build an effective ML-based IDS, it is essential to align detection models with structured, real-world representations of adversary behavior. The MITRE ATT&CK Cloud Matrix offers a widely adopted, standardized knowledge base of tactics and techniques used by threat actors, making it an effective tool for evaluating and enhancing IDS strategies [35].

The MITRE ATT&CK Cloud Matrix is a structured framework that explains how attackers operate inside cloud environments by breaking their behavior into simple, understandable components. It combines malicious actions into high-level tactics such as gaining initial access, stealing credentials, moving through a cloud system, or disrupting services, and pairs each tactic with detailed techniques that show the specific methods attackers use in real cloud platforms [126, 127].

The original development of the MITRE ATT&CK framework for enterprise systems has been extended to include cloud-specific tactics and techniques that provide a unified modeling framework for describing adversarial behavior used by both practitioners and researchers. Prior work on MITRE ATT&CK frameworks has demonstrated value in other domains such as industry settings and adjacent domains. Zhao et al. [128, 129] have created a series of serious games based on MITRE ATT&CK (cloud matrix) elements of the CSA control matrix. These games raised industry awareness of cloud threats and defenses. Xiong et al. [130] also used the MITRE ATT&CK enterprise matrix to build a formal cyber threat model through software and systems modeling techniques. They stressed the importance of structured risk assessment with this type of approach. [131] also demonstrated the applicability of the MITRE ATT&CK cloud matrix framework beyond the enterprise setting. They incorporated their taxonomy of industrial controls into a Bayesian network to model adversary behavior in operational technology systems.

These studies demonstrated the application of the MITRE ATT&CK framework for designing targeted training environments or modeling specific attack scenarios. However, most of these studies focused on limited tactics or narrowly defined subsets of adversary behavior rather than providing a holistic view of threat coverage. Motivated by this, our goal is to provide a more comprehensive perspective by systematically mapping the attacks detected in the literature on ML-based cloud IDS to the full MITRE ATT&CK cloud matrix, as shown in Table 7. This broader mapping highlights which adversarial tactics are well-represented across the research landscape and where significant detection gaps remain.

The majority of the reviewed literature analyzed during this work depends on legacy datasets such as NSL-KDD, KDDCup99, UNSW-NB15, and CICIDS2017/2018 which were developed before the development of the MITRE ATT&CK Cloud Matrix. Therefore, these datasets are not annotated for native ATT&CK techniques. The mappings shown in this work were made using the behavioral intent and operational goals of each type of attack instead of its original label from the dataset. Each type of attack was examined based on its underlying adversary behavior and then matched with the most applicable MITRE ATT&CK Cloud tactics and techniques. For instance, DoS attacks were matched to the Impact tactic since the ultimate goal of DoS attacks is to disrupt services, with host-level DoS behaviors aligned to Endpoint Denial of Service (T1499) and network-level flooding behaviors aligned to Network Denial of Service (T1498). When no direct cloud-specific technique was available, the most behaviorally aligned MITRE ATT&CK Cloud technique that represented the

Table 7 Study-level mapping of attack categories from the reviewed ML-based cloud IDS literature to MITRE ATT&CK cloud matrix

Primary studies	Attack	MITRE ATT&CK cloud matrix techniques	MITRE ATT&CK cloud matrix tactics
[24, 37–41, 43–47, 49, 50, 52, 53, 57, 59, 61, 62, 65, 67–69, 71, 76, 77, 80, 82, 89, 90, 94, 96–98, 100, 101, 103, 106, 107, 115, 116, 118]	DoS-Based Attacks	Endpoint DoS (T1499); Network DoS (T1498)	Impact
[37, 38, 41, 43, 47, 49, 50, 65, 67–69, 71, 76, 82, 89, 90, 93, 94, 96–98, 100, 103, 106, 116]	R2L (Remote-to-Local)	Vary with respect to technique (Exploit Public Facing Application (T1190)/Brute Force (T1110))	Initial Access/ Credential Access
[37, 38, 41, 43, 47, 49, 50, 65, 67–69, 71, 76, 82, 89, 90, 93, 94, 96–98, 100, 103, 106, 116]	U2R (User-to-Root) Attacks	Abuse Elevation Control Mechanism (T1548)	Privilege Escalation
[37, 40, 44–46, 49, 52, 57, 77–79, 93, 98, 118]	Shellcode Exploits	Exploitation for Defense Evasion (T1211)	Defense Evasion
[37, 39, 40, 44–46, 49, 52, 57, 77–79, 93, 118]	Other Exploit-Based Attacks	Exploitation for Client Execution (T1203); Exploit Public-Facing Application (T1190)	Execution; Initial Access
[37, 39, 40, 44–46, 49, 52, 57, 71, 76–79, 93, 98, 118]	Reconnaissance-Based Attacks (Port Scanning, Fuzzers, Profiling)	Network Service Discovery (T1046); Account Discovery (T1087); System Discovery (T1082)	Discovery
[37, 39, 40, 44–46, 49, 52, 57, 71, 76–79, 93, 98, 118]	Backdoor Attacks	Impair Defenses (T1562)	Defense Evasion
[37, 39, 40, 44–46, 49, 52, 57, 76–79, 93, 98, 118]	Worm-Based Attacks (cloud worms)	Remote Services (T1021)	Lateral Movement
[37, 52, 59, 61, 71, 80, 94, 107, 118]	Brute-Force Attacks	Brute Force (T1110)	Credential Access
[37, 39, 40, 44–46, 49, 52, 57, 71, 76–79, 93, 98, 118]	Exfiltration (FTP and HTTP)	Exfiltration Over Alternative Protocol (T1048); Exfiltration Over Web Service (T1567)	Exfiltration
[37, 52, 53, 59, 61, 71, 80]	Infiltration Attacks (Phishing, Account Hijacking)	Phishing (T1566); Account Hijacking (T1078)	Initial Access
[58, 62, 102, 115, 118]	SQL Injection	Exploit Public-Facing Application (T1190)	Initial Access
[58, 118]	Cross-Site Scripting (XSS)	Drive-By Compromise (T1189)	Initial Access
[41, 58, 71, 118]	SSH-Based Attacks	Remote Services (T1021)	Lateral Movement
[119]	Hyperjacking Attacks	Command & Scripting Interpreter (T1059); Cloud Administration Command (T1651)	Execution
[66, 75, 81]	Resource-Based Attacks	Resource Hijacking (T1496)	Impact
[62, 71]	Heartbleed Attack	Exploitation for Credential Access (T1212)	Credential Access
[62]	Authorization Bypass Attacks	Abuse Elevation Control Mechanism (T1548)	Privilege Escalation
[62, 119]	Container Escape, VM Escape Attacks	Escape to Host (T1611)	Privilege Escalation
[85–88]	Adversarial Evasion Attacks (FGSM, PGD, DeepFool, Square)	Impair Defenses (T1562)	Defense Evasion
[119]	Side-Channel Attacks	Unsecured Credentials (T1552)	Credential Access
[73]	Inference Attacks	Account Discovery (T1087)	Discovery

same behavior was used. For example, adversarial evasion attacks were mapped to Impair Defenses (T1562) because their objective is to bypass or mislead the intrusion detection mechanism. This adopted approach enables a consistent interpretation of legacy benchmark attacks within a contemporary cloud-security threat model.

This mapping provides the following key benefits:

- This reveals that MITRE ATT&CK Cloud Matrix tactics and techniques are frequently addressed in existing ML-based IDS research.
- This highlights the limitations of the current literature, where certain adversarial behaviors remain underexplored or undetected.
- It supports the standardization and alignment of academic contributions with real-world, industry-recognized threat modeling frameworks.

The primary objective behind developing this mapping was not to assess the inherent detection capabilities of various ML-based IDS architectures on their own merit based upon how well the training and testing datasets represent them. The primary function of the mapping is as a literature-based evaluation of the current state of cloud IDS research that is ML-based. Within this scope, "coverage" represents the extent to which the specific attacks studied in each selected study, along with those respective studies' datasets, labels for these attacks, sources for cloud telemetry data, and the types of attacks being detected by the ML-based IDS models, align with the tactics and techniques, and/or tactics and techniques described within the MITRE ATT&CK Cloud Matrix. If a particular tactic is either mapped poorly or not at all, it is not necessary to assume that the ML-based IDS models cannot detect that type of behavior. In contrast, this indicates that this kind of behavior has either not been included, assessed, or reported within current research on cloud IDS. The difference here is significant as the gaps identified could be due to the limitations of the datasets; lack of attack labels; inadequate cloud data collection; specific design decisions made by model developers; or larger issues with how primary studies were performed. The following paragraphs present the mapping methodology used in this study.

4.1.2 Mapping methodology

To perform the mapping, we manually extracted the primary attack types addressed in each reviewed study in Sect. 3 and aligned them with the corresponding MITRE ATT&CK Cloud Matrix tactics and techniques. The MITRE ATT&CK cloud framework describes tactics as *"what the attacker is trying to achieve"*, such as getting in, running code, gaining higher privileges, or stealing data. These are labeled with "TA" codes. Techniques explain *"how the attacker actually does it"*, and the specific method or action used is marked with "TA" codes.In short, tactics show an attacker's goal, and techniques show the exact way they carry it out [35]. The mapping focused on the *main intent of the attacks*. Table 7 lists the details of the attacks and their corresponding alignments, whereas Fig. 11 illustrates the overall mapping. In order to clearly show how these studies map to the MITRE ATT&CK Cloud Matrix (Table 7), the table includes all relevant data related to an individual primary study that was included in this review, as well as all applicable MITRE tactics and techniques. Each row is based on a common set of studies that reported similar attack behavior and were therefore mapped to the same MITRE tactic and techniques group. When a study addressed more than one attack category, it may appear in more than one row within the table. Similarly, when multiple studies addressed the same attack behavior, they were grouped under a single row to avoid repetition while preserving traceability between the reviewed papers and the MITRE mapping. Therefore, the tactic-level discussion provided in the following section should be read together with Table 7, which references all the relevant study-level evidence used to map the various attack categories, techniques, and tactics from this review. In the figure, the techniques shaded in gray denote the mapped associations, whereas the white cells indicate techniques with no identified alignment.

Initial access: Initial Access is an attempt to gain unauthorized entry into the cloud environment. There are various means to gain initial access to the cloud resources. One method mentioned in the primary studies is SQL injection [58, 62, 102, 115, 118]. SQL injection is an attack in which an attacker inserts malicious SQL commands into an application's input fields to access or manipulate the database illegally. SQL injection is directly linked to *exploit public-facing applications (which is identified by the code (T1190))* because the attacker breaks in through a web application that's openly accessible on the internet. They gain their first unauthorized entry through the injection of malicious queries into the exposed interface; hence, it falls under "Initial Access." Similarly, Cross-Site Scripting (XSS) [58, 118] has been one area of interest for researchers. XSS involves an attacker injecting malicious scripts onto trusted websites in order to steal data, hijack sessions, or alter how users interact with content. XSS fits well within "Drive-By Compromise," identified by code (T1189), as the attacker injects malicious scripts onto pages loaded by the victim. Once the scripts run in the victim's browser, the attacker gains an initial foothold into the system without having previously accessed it. Phishing and account hijacking infiltration attacks [37, 52, 53, 59, 61, 71, 80] correspond to T1566 (Phishing) and T1078 (Valid Accounts); thus, they are classified as "Initial Access" attacks. The studies that primarily relate to these Initial Access attack categories and their associated MITRE ATT&CK Cloud Matrix techniques are presented in Table 7.The highlighted techniques in Fig. 11 under the tactic "Initial Access" show the mapping of these attacks.

Execution: Execution is a method of executing bad code or attempting to execute malicious commands within cloud services/workload. Hyper-jacking attacks [119] describe attempts to "take over" the hypervisor layer that controls multiple VMs. This can be done without the knowledge of the user. These attacks fall into two categories of techniques, due to the fact that the attackers issue their malicious commands via both common hypervisor interfaces and ESXi-specific administrative functions. As such, the first category falls under the technique "Command and Scripting Interpreter" (T1059). More specifically, this falls under the sub-category

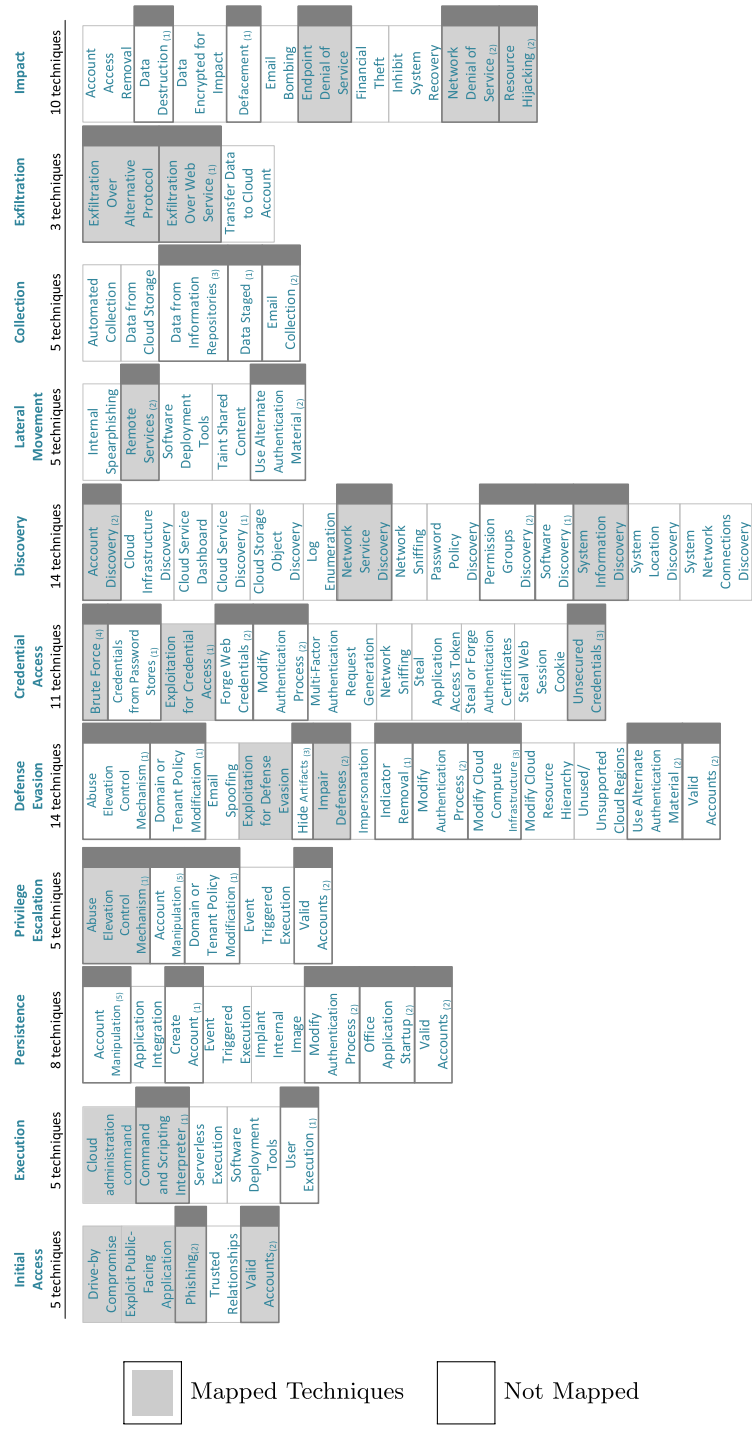


Fig. 11 Alignment of attacks addressed with the MITRE ATT&CK Cloud Matrix

"Hypervisor CLI" (T1059.012) since they use the Hypervisor's CLI to issue their malicious commands. Similarly, the second category falls under "Cloud Administration Command" (T1651), specifically the sub-technique "ESXi Admin Command," since they have issued their malicious commands via the ESXi admin interface. Through the use of the Hypervisor's CLI, or the ESXi admin interface, the attacker has loaded rogue modules onto the system, altered VM states, or taken control of the VM. In doing so, the attacker was able to run commands/scripts to take control of the VMs in question. Therefore, it maps to Cloud Administrative Commands at the time of execution. According to MITRE, these types of techniques fall under the Tactic "Execution". The mapping results are shown in Fig. 11.

Persistence: Persistence refers to attempts to maintain long-term access using cloud accounts, tokens, or

configuration changes. *Across all primary studies, no attack type directly corresponds to MITRE's "Persistence" tactic.* Accordingly, no primary study-to-technique mapping is reported for this tactic in Table 7.

Privilege escalation: Attempts to obtain elevated permissions such as admin, root, or privileged IAM roles. U2R attacks (which occur when a normal user finds a flaw in the system and uses it for full root-level control) fall under technique T1548 (T1548 is when an attacker misuses built-in elevation mechanisms to gain higher-level permissions), which is mapped to the abuse elevation control technique, the exact behavior defined by MITRE as privilege escalation. Authorization Bypass attacks [62] also map to T1548, because bypassing access controls gives the attacker higher permissions than they should have, directly matching MITRE's elevation-control abuse techniques, and can further lead to account manipulation. VM Escape and Container Escape attacks [62, 119] also fit this category, as breaking out of an isolated VM or container into the host environment requires obtaining privileges far beyond what the environment originally grants. The primary studies corresponding to these privilege-escalation attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7.

Defense Evasion: Attempts to avoid detection by disabling logs, bypassing monitoring, or hiding activities in cloud systems. Shellcode exploits fit primarily under "Defense Evasion" as the attacker uses them to bypass security controls, letting their code run without being blocked or noticed, and fall under "Exploitation for Defense Evasion (T1211)". A backdoor attack (occurs when an attacker plants a hidden way to access a system without going through normal security checks) also aligns with the "Defense Evasion" under technique "Impair Defenses" (T1562). Moreover, adversarial attacks such as FGSM, PGD, DeepFool, and Square [85–88] fall directly under MITRE ATLAS, which focuses on adversarial machine learning techniques [132]. In the standard MITRE ATT&CK Cloud Matrix, they can only be aligned with "Defense Evasion" (Impair Defenses T1562) because the objective of adversarial examples is to hide malicious activities by misleading the IDS. The primary studies corresponding to these Defense Evasion attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7. The highlighted techniques shown in Fig. 11 under tactic "Defense Evasion" display these attacks.

Credential access: Attempts to steal cloud passwords, API keys, tokens, or authentication secrets. A brute-force attack (occurs when an attacker repeatedly tries different combinations of passwords or credentials until they find the correct one and gain access) falls under "Credential Access" (T1110), directly aligns with the technique "Brute Force" because the attacker repeatedly guesses passwords to obtain valid credentials. Side-channel attacks [119] (steal sensitive information by observing a system's unintended leaks, such as timing or power patterns) align with the technique "Unsecured Credentials" (T1552) because they extract keys or tokens from weakly protected sources or leaked system signals. Heartbleed attack [62, 71] (exploits vulnerabilities in OpenSSL to read sensitive data directly from a server's memory) directly maps to "Exploitation for Credential Access" (T1212) because it leaks sensitive memory containing passwords, tokens, and private keys. All these methods ultimately aim to steal authentication credentials. Therefore, we map them to the "Credential Access" tactic. The primary studies corresponding to these Credential Access attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7. The techniques highlighted in Fig. 11 represent these types of attacks.

Discovery: Attempts to identify cloud resources, services, identities, roles, and network configurations. Reconnaissance-based attacks (attacks related to early-stage attempts where an attacker quietly scans and gathers information about a system to find vulnerabilities for attack later). For example, port scanning and fuzzing fall under "Network Service Discovery" (T1046) because they probe open ports, services, and system details to understand the environment before any exploitation, whereas an environment-profiling attack is directly mapped to "System Discovery" (T1082) because the attacker's main goal is to analyze how the cloud environment is built. Similarly, inference attacks [73] directly align with "Account Discovery" (T1087) because they analyze system responses or behavior to identify valid usernames and account structures. Both attack types are purely information-gathering steps, which is why we map them to the "Discovery" tactic in Fig. 11. The primary studies corresponding to these Discovery attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7.

Lateral movement: Attempts to move across cloud workloads, accounts, or services using compromised permissions. Worm-based attacks (cloud worms) fit this tactic as they spread automatically across cloud instances by abusing remote-access channels that are already open between VMs. SSH-based attacks [41, 58, 71, 118] also validate this mapping because SSH is a remote-access protocol. When an attacker uses SSH to log into a machine after cracking the credentials, it directly matches T1021. All these behaviors enable the attacker to move beyond the entry point and access additional systems, which is the core purpose of the "Lateral Movement" tactic. The primary studies corresponding to these Lateral Movement attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7. We highlighted "Remote Services" in Fig. 11 under the "Lateral Movement", as these attacks use remote access paths to gain access into other components of the environment.

Collection: Attempts to gather data from cloud storage, SaaS applications, databases, and workloads. None of the primary studies address any attack that directly maps to MITRE's "Collection" tactic because benchmark IDS datasets do not include post-compromise data-gathering behaviors such as cloud storage access, email collection, or internal data staging. Accordingly, no primary study-to-technique mapping is reported for this tactic in Table 7.

Exfiltration: Attempts to transfer cloud data outside the environment through APIs or external endpoints. FTP-based exfiltration (this attack steals data by quietly sending files out through an FTP connection). It maps to "Exfiltration Over Alternative Protocol" (T1048), because FTP is a non-web protocol used as a separate outbound channel. Attackers use it to slip data out through an older, less-monitored path. HTTP-based exfiltration (This attack hides stolen information inside normal HTTP/HTTPS requests). It maps to "Exfiltration Over Web Service" (T1567) because it uses standard web traffic to disguise data theft. Therefore, these two attacks are mapped purely under the tactic "Exfiltration" with technique "Exfiltration Over Alternative Protocol" and "Exfiltration Over Web Service", as highlighted in Fig. 11. The primary studies corresponding to these Exfiltration attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7.

Impact: Attempts to disrupt, degrade, or damage cloud workloads, services, and data availability. DoS-based attacks interrupt a system or service with excessive traffic, so it cannot respond to legitimate users. These attacks directly map to "Endpoint DoS" (T1499) in the case of host-based denial-of-service attacks and "Network DoS" (T1498) in the case of network-level denial-of-service attacks. These attacks directly interrupt the cloud network, causing service disruption as defined under MITRE's "Impact" tactic. Resource-based attacks [66, 75, 81], in which an attacker secretly uses the victim's cloud or resources for their benefit, match the technique "Resource Hijacking" (T1496), which also fits here because abnormal CPU, memory, or I/O consumption leads to the same availability loss and operational degradation as DoS behavior. The primary studies corresponding to these Impact attack categories and their mapped MITRE ATT&CK Cloud Matrix techniques are listed in Table 7.

The studied attack types were related to the MITRE ATT&CK Cloud Matrix based upon direct behavioral correlation with the best corresponding cloud tactics and techniques. These correlations result in each of the attack types being associated with only those techniques which reflect their actions or intents. In addition to providing many insights into both trends and limitations in current machine learning (ML)-based intrusion detection systems (IDS) research for cloud computing environments, the correlations found in the study indicate that certain cloud-related tactics within the MITRE ATT&CK Cloud Matrix, i.e., *Impact*, *Execution*, and *Initial Access*, have been frequently reported by previous researchers. Conversely, other cloud-related tactics, including *Persistence* and *Collection*, receive no attention across the reviewed studies. This same lack of representation has similarly impacted the scope of the dataset studied. Not only did it contain stale network traffic patterns; it also lacked attack phases specific to cloud computing environments. Therefore, future IDS benchmarks specifically designed to support cloud computing environments will need to include attack sequences supporting previously unaddressed cloud computing tactics such as *Persistence* and *Collection*, as discussed in greater detail in Observation 5.

4.2 Gap analysis

This section summarizes the gaps and observations extracted through a systematic analysis of the studies discussed in Sect. 3.2 and their mapping to the MITRE ATT&CK Cloud Matrix in Sect. 4.1. These observations indicate the shortcomings identified and outline areas where further investigation is needed to design efficient intrusion detection systems for cloud environments.

Observation 1:*Absence of Cloud-Native Threats in Benchmark DatasetsMeta-analysis:* A significant research gap exists owing to the attack types that remain unaddressed in current benchmark datasets. Many modern, cloud-native threats, such as cloud storage misconfigurations (e.g., exposed S3 buckets), function hijacking, cloud metadata service exploitation, Virtual Machine Introspection (VMI) bypasses, and multi-tenant attacks, are not present in the datasets that are commonly used to train and evaluate current IDS models. Consequently, the scope of detectable threats is constrained, and numerous MITRE ATT&CK Cloud Matrix tactics and techniques remain unexplored. This loophole indicates a dire need for a comprehensive cloud-specific dataset that can be used to develop effective IDS models capable of detecting advanced threats.

Observation 2:*Under-representation of MITRE ATT&CK Cloud Tactics in Existing IDS Evaluations*

Meta-analysis: Another gap identified through the mapping is that nearly all of the existing ML-based cloud IDS studies have been based upon a limited number of common types of attacks (DDoS, brute force, and privilege escalation) that closely match the MITRE ATT&CK Cloud Matrix tactics for Impact, Initial Access, and Privilege Escalation. As shown in Fig. 11, much less attention has been paid to the sophisticated cloud-related behaviors associated with Persistence and Collection. The disparity in attention toward these two areas is not indicative that ML-based IDS models will not be able to identify these behaviors if sufficient data and telemetry exist. Rather, it illustrates that there exists an unbalanced representation of these behaviors within the datasets, attack labels, telemetry sources, and evaluation settings used by the researchers

whose work was included in this study. Thus, future research efforts would benefit from addressing both the creation of new cloud-native datasets and the collection of realistic cloud telemetry. Additionally, developing attack labels at a finer granularity than what most current studies employ and developing IDS models capable of generalizing beyond specific test cases would provide substantial value to the development of effective cloud IDS systems.

Observation 3:*Prevalence of Supervised Learning Over Unsupervised ApproachesMeta-analysis:* An additional major void was identified through the evaluation conducted for this systematic review; specifically, most studies we analyzed relied heavily upon pre-labeled data. A statistical assessment of our findings indicated that an approximate total of 74.7 percent of the studies included within our systematic review employed a type of supervised machine learning methodology, while less than four percent (specifically, 3.8 percent) used unsupervised machine learning methodologies. Ultimately, these results indicate that there has been considerable attention paid to identifying labeled and known classes of attack within studies employing ML-based IDS models. It is important to note that these results do not signify that ML-based IDS models cannot identify zero-day or unknown threats. Rather, they demonstrate that little methodological effort has been made in developing IDS models that can identify such threats within the reviewed body of literature relating to cloud-based IDS. Since zero-day and emerging attacks may not be adequately represented by the fixed labels used in existing benchmark datasets, future cloud IDS research must place greater emphasis on unsupervised, semi-supervised, self-supervised, open-set, and continual-learning methods that can identify novel, evolving, or previously unseen attack behaviors.

Observation 4:*Lack of Standardized IDS Solutions for Multi-Cloud DeploymentsMeta-analysis:* As the industry increasingly adopts multi-cloud architectures, the need for standardized and interoperable IDS frameworks across heterogeneous cloud environments has become critical. However, our review reveals a significant gap: the majority of existing studies have focused solely on single-cloud deployments. IDS solutions tailored for multi-cloud or hybrid environments remain underexplored, limiting the applicability of the current research in real-world distributed cloud settings.

Observation 5:*Reliance on legacy or Simulated Datasets Limiting Benchmark ComparabilityMeta-analysis:* An important limit of many current ML-based Cloud IDS studies remains that most of them continue to depend upon data from a past era, whether it be an older dataset or one that was designed for another type of network architecture. The majority of the literature currently utilizes legacy datasets like KDDCup99 and NSL-KDD (which can be seen in Fig. 10),which represent neither the complex threat environment nor the true nature of today's cloud networks. In addition, there are numerous examples of researchers creating their own custom or simulated datasets, therefore reducing consistency across various types of traffic and making comparison of benchmarks or results across multiple studies difficult. To overcome shortcomings of legacy IDS benchmarking datasets, we recommend developing future cloud-centric IDS benchmarking datasets based on specific, cloud-specific IDS benchmark dataset features (aligned with both the MITRE ATT&CK Cloud Matrix and the gap in coverage shown in Fig. 11).

First, such datasets should contain cloud-native attack cases not present in either KDDCup99 or NSL-KDD, namely: VM escape, container escape, hyperjacking, side-channel attacks, use of cloud metadata services for exploit, misuse of exposed cloud storage, abuse of serverless functions, and hijacking of cloud accounts. *Second*, the dataset should include explicit coverage of MITRE-defined Tactics currently unmapped or poorly represented in previous IDS evaluations (i.e., Persistence and Collection). For Persistence, example behaviors for benchmark traces include: account manipulation, creating unauthorized cloud accounts, illegitimate use of valid cloud accounts, event-triggered execution, modifying tenant/domain policy, and implantation of malicious images into internal cloud infrastructure. For Collection, examples of behaviors for this category include: automated collection, collecting data from cloud storage, collecting data from information repositories, collecting data via email/SaaS platforms, and staging of collected data before exfiltration.

Third, the dataset should include multi-stage attack chain examples instead of single-stage attack examples, to enable analysis of the connected sequence of Initial Access, Execution, Persistence, Privilege Escalation, Discovery, Collection, Exfiltration, and Impact. *Fourth*, the dataset should also include multi-layered telemetry types, including but not limited to: network flows, host logs, system call traces, hypervisor events, cloud audit logs, IAM activity logs, API call traces, container/Kubernetes logs, storage-access logs, and serverless execution traces.

Fifth, each entry in the dataset should be annotated at multiple levels (benign vs. malicious classification; attack type/family; attack stage/MITRE tactic/MITRE technique; timestamp; affected cloud layer(s); and source of telemetry). *Sixth*, the dataset should include realistic benign workloads and background noise from public, private, hybrid, and multi-cloud environments to avoid overfitting to synthetic traffic patterns. *Finally*, the dataset should provide reproducible generation procedures, clear documentation of all attack scenarios included, class-imbalance information, and predefined evaluation splits for known-attack detection, zero-day detection, cross-cloud generalization, and temporal validation.

Observation 6:*Insufficient Attention to Adversarial AttacksMeta-analysis:* Adversarial attacks are significantly increasing in modern cloud environments. The effectiveness

of an IDS in the real world depends largely on its resilience against adversarial attacks. However, our analysis reveals that only a limited number of studies have explicitly addressed adversarial robustness in cloud-based IDS. This indicates a significant research gap in the development of detection models capable of addressing evasion, poisoning, or other adversarial tactics.

Observation 7:*Research Bias Toward NIDS over HIDSMeta-analysis:* Our investigation reveals a significant imbalance in IDS deployment focus, and the majority of existing research emphasizes network-based IDS (NIDS), while host-based IDS (HIDS) remains underexplored, as shown in Fig. 9. Since cloud environments extensively rely on virtualization, containers, and host-level operations, effective monitoring at the host level is essential. The limited attention to HIDS in contrast to the extensive focus on NIDS highlights a critical gap in ensuring comprehensive cloud security coverage.

Observation 8:*Feature-Dependent ML Models and the Potential of LLM-Driven DetectionMeta-analysis:* Since we have observed that most ML-based IDS approaches rely excessively on predefined feature representations, their ability to detect unknown or evolving threats is limited. This lack of flexibility reduces the system's ability to respond effectively to emerging attack patterns. Traditional ML-based IDS architectures typically rely upon predefined numerical features, pre-defined attack types (labels), and dataset-specific processing rules. Due to these limitations, traditional ML-based IDS architectures often lack adaptability to novel combinations of API calls, IAM activities, audit logs, container events, system logs, or multi-stage behaviors that have not been encountered during model training.

LLMs are an attractive alternative due to their ability to semantically interpret various forms of unstructured and structured security data (e.g., cloud audit trails, IAM events, API calls, system logs, alerts, vulnerability descriptions, analyst reports). This capability can help correlate events across multiple cloud layers, identify suspicious behavioral patterns, generate contextual explanations, map observed activities to MITRE ATT&CK tactics and techniques, and support weak-label generation for poorly labeled or emerging attack scenarios. Thus, instead of relying solely on static feature vectors, LLM-enabled IDS architectures can dynamically generate abstractions based on context from evolving cloud telemetry.

However, LLMs are not meant to replace validated ML/DL-based IDS architectures or benchmarks. Outputs generated by LLMs may be influenced by hallucination, insufficient context, sensitivity to prompts, and deployment costs. Future LLM-enabled cloud IDS should therefore be developed using a layered approach including supervised reasoning and decision-making capabilities; combined with traditional ML-based IDS architectures, anomaly detection, and rule-based validation; supplemented by expert feedback and validation against MITRE-defined frameworks. Thus, LLMs can augment IDS decision-making reliability via adaptive detection and increased explainability.

5 Conclusions and future work

This comprehensive study emphasizes the effectiveness of an ML-based IDS specifically for cloud environments. Following PRISMA guidelines, we systematically investigated various recent studies in the selective domain. The main objectives of this study were to present the current landscape of ML-based IDS for cloud environments and identify the gaps between cloud IDS academic research and industrial security standards. To achieve this, we first categorized the primary studies based on our proposed taxonomy, which organized the literature into three major categories and enabled an examination of trends across IDS architectures, learning models, feature extraction approaches, dataset usage, and evaluation metrics. The next step was to align the attacks included in these studies with the industry standard MITRE ATT&CK Cloud Matrix. This approach not only provided better insights into the industrial relevance of the reviewed studies but also enabled the identification of several critical gaps, including unaddressed attack vectors. The observations from this review offer key directions for the advancement of future IDS solutions that should focus on developing cloud-native datasets, adversarially robust and lightweight systems, as well as multi-cloud compatible IDS solutions that are more efficient, secure, and better aligned with real-world threat landscapes. Moreover, the role of LLMs in designing IDS solutions cannot be overlooked in the future because of their contextual reasoning ability, which will make them more effective for securing cloud environments.

Appendix A quality assessment scores of selected primary studies

This appendix is generated based on the five QA criteria defined in Sect. 2.5. The assessment of the studies for each criterion is based on a three-point scale: 1) 0 means not satisfied, 2) 0.5 means partially satisfied, and 3) 1 indicates the criterion is fully satisfied. Since five quality assessment criteria are defined, the maximum score a study can achieve is 5. Each distinct study is assessed only once here to avoid duplicate scores.

See Table A1.

Table A1 Quality assessment of the 79 primary studies. Each criterion (QA1-QA5) is scored 0/0.5/1; H = High, M = Medium, L = Low

Reference	Study	QA1	QA2	QA3	QA4	QA5	Total	Rating
Performance Optimization → Nature-Inspired								
[57]	Manzoor et al. 2023	1	1	1	1	0.5	4.5	H
[47]	Sanagana & Tummalachervu 2024	1	1	0.5	1	0.5	4	H
[37]	Maheswari et al. 2023	0.5	1	1	1	1	4.5	H
[38]	Silambarasan et al. 2024	1	1	1	1	0.5	4.5	H
[39]	Nagamani et al. 2025	1	1	1	1	0.5	4.5	H
[40]	Bajpai & Patankar 2024	1	1	1	0.5	1	4.5	H
[41]	Jain et al. 2023	1	1	1	1	1	5	H
[43]	Parkash & Mittal 2023	1	1	1	0.5	0.5	4	H
[44]	Sarkar et al. 2023	1	1	1	0.5	0.5	4	H
[48]	Eswari et al. 2023	1	1	0	1	0	3	M
[58]	Tiwari & Murugappan 2023	1	1	1	1	0.5	4.5	H
[59]	Rana et al. 2024	1	1	1	1	1	5	H
[49]	Sreelatha et al. 2022	1	1	1	0.5	1	4.5	H
[50]	Jansi Sophia Mary & Mahalakshmi 2024	1	1	1	1	0.5	4.5	H
[51]	Ravindranath et al. 2020	0.5	1	0.5	0.5	0.5	3	M
[45]	Arunkumar & Kumar 2023	1	1	1	0.5	0.5	4	H
[46]	Bakro et al. 2024	1	1	1	1	1	5	H
[52]	Radhika 2024	1	1	1	1	1	5	H
[53]	Bai & Punithavalli 2024	1	1	1	1	0.5	4.5	H
[60]	Attou et al. 2023	0.5	1	1	1	1	4.5	H
Performance Optimization → Lightweight								
[61]	Kim et al. 2024	1	1	1	0.5	1	4.5	H
[62]	Joraviya et al. 2024	1	1	1	1	0.5	4.5	H
[63]	Liu et al. 2020	1	1	1	0	0.5	3.5	H
[64]	Gill et al. 2024	1	1	1	0.5	1	4.5	H
[65]	Ghosh et al. 2022	1	1	1	0.5	1	4.5	H
[66]	Mitropoulou et al. 2023	1	1	0.5	1	0.5	4	H
[67]	Sethi et al. 2020 (SPACE)	1	1	1	0.5	1	4.5	H
[68]	Bakro et al. 2022	1	1	1	1	1	5	H
[69]	Gupta & Alam 2024	1	1	1	1	0.5	4.5	H
Learning and Adaptive → Federated Learning								
[71]	Ayed & Talhi 2021	1	1	1	0.5	0.5	4	H
[72]	Luo & Ji 2025	0.5	1	0.5	0.5	0.5	3	M
[73]	Tyagi & Sharma 2025	0.5	1	0.5	1	0.5	3.5	H
Learning and Adaptive → Anomaly Detection								
[74]	Bhingarkar et al. 2023	0.5	1	1	0.5	1	4	H
[75]	Jiang 2024	1	1	0.5	1	0.5	4	H
[76]	Zuhair et al. 2022	1	1	1	1	1	5	H
[77]	Parameswarappa et al. 2023	1	1	1	1	0.5	4.5	H
[78]	Chkirbene et al. 2021	1	1	1	1	0.5	4.5	H
[79]	Shahzad et al. 2022	1	1	1	1	0.5	4.5	H
[80]	Al-Ghuwairi et al. 2023	1	1	1	0	0.5	3.5	H
[81]	Xin et al. 2023	0.5	1	0.5	0.5	1	3.5	H
[82]	Althoubi & Peyravi 2023	1	1	1	1	0.5	4.5	H
[83]	Sethi et al. 2020 (COMSNETS)	1	1	1	0.5	0.5	4	H
[84]	Attou et al. 2023 (BDMA)	0.5	1	1	0.5	1	4	H
Learning and Adaptive → Adversarial-Resilient								
[85]	Holla et al. 2025	1	1	1	0.5	0.5	4	H
[86]	Patel et al. 2025	0.5	1	0	0	0	1.5	L
[87]	Peng et al. 2020	1	1	1	0	0.5	3.5	H
[88]	Ganesan & Sarac 2021	1	1	1	0.5	1	4.5	H
Learning and Adaptive → Traditional ML								
[24]	Vu et al. 2023	1	1	1	0	1	4	H
[89]	Hizal et al. 2021	1	1	1	0.5	0.5	4	H

Table A1 (continued)

Reference	Study	QA1	QA2	QA3	QA4	QA5	Total	Rating
[90]	Geetha et al. 2024	1	1	1	1	1	5	H
[91]	Venkateshwarlu et al. 2023	1	1	1	0.5	0.5	4	H
[92]	Srilatha & Shyam 2021	1	1	1	0.5	0.5	4	H
[93]	Bakro et al. 2023	1	1	1	1	1	5	H
[94]	Singh & Ranga 2021	1	1	1	0.5	0.5	4	H
[95]	Daoud et al. 2021	1	1	0.5	1	0.5	4	H
[96]	Khan & Haroon 2023	1	1	1	1	0.5	4.5	H
[97]	Shukla & Sharma 2023	1	1	1	0.5	1	4.5	H
[98]	Jiang et al. 2023	1	1	1	1	1	5	H
[99]	R. et al. 2024	1	1	1	0.5	0.5	4	H
[100]	Kassimi et al. 2022	1	1	1	0	0.5	3.5	H
[101]	Fugkeaw et al. 2023	1	0.5	0.5	0.5	0.5	3	M
[102]	Tripathy et al. 2020	1	1	0.5	1	0.5	4	H
[103]	Onyema et al. 2022	1	1	1	1	0.5	4.5	H
[104]	Veena et al. 2022	0.5	1	0	0.5	0	2	L
[105]	Wang 2023	1	1	0.5	1	0.5	4	H
[106]	Krishnaveni et al. 2021	1	1	1	1	1	5	H
[107]	Kandakatla & Rajakumari 2024	1	1	0.5	1	0.5	4	H
[108]	Kumar & Sharma 2023	1	1	0.5	1	0.5	4	H
[109]	Sharma et al. 2023	1	1	0.5	0.5	1	4	H
[110]	Jisna et al. 2021	1	1	1	1	0.5	4.5	H
Environment-Aware → Multi-Cloud								
[115]	Srinivasulu et al. 2024	1	1	0.5	0	1	3.5	H
[116]	Chaudhary et al. 2024	1	1	1	0.5	1	4.5	H
[117]	J. et al. 2023	1	1	0.5	0	0.5	3	M
[118]	Nizamudeen 2023	1	1	1	0.5	1	4.5	H
Environment-Aware → Hypervisor								
[54]	Jaber & Rehman 2020	1	1	1	1	0.5	4.5	H
[119]	Alazab et al. 2025	1	0.5	1	1	1	4.5	H
[120]	Anumukonda et al. 2024	0.5	1	1	1	1	4.5	H
[121]	Kumar et al. 2025	1	1	1	0.5	1	4.5	H
[122]	Qaffas 2024	1	1	0.5	1	0.5	4	H

Acknowledgements The authors acknowledge the support from the University of Waikato, Hamilton, New Zealand.

Author contributions All authors contributed equally to this work and reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Merlo, T.R., Fard, F., Hawamdeh, S.: Cloud computing's impact on the digital transformation of the enterprise: a mixed-methods approach. Sustainability **17**(13), 5755 (2025). https://doi.org/10.3390/su17135755
2. Filiopoulou, E., Dede, G., Evangelatos, S., Kamalakis, T., Michalakelis, C., Nikolaidou, M.: The economic dimensions of cloud security: a systematic literature review. Comput. Secur. 105139 (2026)
3. Alam, M., Shahid, M., Mustajab, S., Sajid, I., Salah, E.G.: Security challenges for workflow allocation model in cloud computing environment: a comprehensive survey, framework, taxonomy, open issues, and future directions. J. Supercomput. **80**(8), 11491–11555 (2024). https://doi.org/10.1007/s11227-023-05873-1
4. Khajeh-Hosseini, A., Greenwood, D., Smith, I., Sommerville, I.: The rise of cloud computing: data protection, privacy and pay-as-you-go models. Front. Big Data **5**, 919765 (2022). https://doi.org/10.3389/fdata.2022.919765

5. Jangjou, M., Sohrabi, M.K.: A comprehensive survey on security challenges in different network layers in cloud computing. Arch. Comput. Methods Eng. **29**(6), 3587–3608 (2022). https://doi.org/10.1007/s11831-022-09708-9
6. Younis, R., Iqbal, M., Munir, K., Javed, M.A., Haris, M., Alahmari, S.: A comprehensive analysis of cloud service models: Iaas, paas, and saas in the context of emerging technologies and trend. In: 2024 International Conference on Electrical, Communication and Computer Engineering (ICECCE), pp. 1–6 (2024). https://doi.org/10.1109/ICECCE63537.2024.10823401
7. Mitchell, B.S.: On the automatic identification of misconfiguration errors in cloud native environments. In: Proceedings of the 2024 ACM Symposium on Cloud Computing (SoCC '24). Association for Computing Machinery, New York, NY, USA (2024). https://doi.org/10.1145/3719384.3719463
8. Qualysec: The top 10 cloud vulnerabilities. (2025). Accessed: Sept. 15, 2025. https://qualysec.com/the-top-10-cloud-vulnerability/
9. Cloud Security Alliance: Top threats to cloud computing: Deep dive 2025. Technical report, Cloud Security Alliance (2025). Accessed: September 18, 2025. https://cloudsecurityalliance.org/artifacts/top-threats-to-cloud-computing-2025
10. Alvarez, M., Carid, C., Chung, J., Cunningham, S., Epley, M., Goyal, M., Hammond, C., Hill, S., Kuo, J., McMillen, D., Mühr, G., Parham, G., Zeizel, A.: Ibm x-force 2025 threat intelligence index. Res. Insights, IBM Red Hat. (2025). https://www.ibm.com/thought-leadership/institute-business-value/en-us/report/2025-threat-intelligence-index
11. Wired: The anatomy of the snowflake breach and its impact. Technical report, Wired (2024). [Online; Accessed 10-March-2025]. https://www.wired.com/story/connor-moucka-snowflake-hack-arrest-extradition
12. Cloud, G.: Cloud resource hijacking: How threat actors reuse decommissioned resources. Technical report, Google (2024). [Online; Accessed 10-March-2025]. https://cloud.google.com/blog/topics/threat-intelligence/unc5537-snowflake-data-theft-extortion
13. News, T.H.: New ALBeast Vulnerability Exposes Thousands of AWS Applications. [Online; Accessed 10-March-2025] (2024). https://thehackernews.com/2024/08/new-albeast-vulnerability-exposes.html
14. News, A.: U.S. Treasury data breach: investigation into API key misuse by State-Sponsored Hackers. [Online; Accessed 10-March-2025] (2024). https://apnews.com/article/8942106afabeac96010057e05c67c9d5
15. Nirmata: Preventive security vs detection and response. https://nirmata.com/2024/03/04/preventive-security-vs-detection-and-response/ (2024). Accessed: 2025-09-21
16. Liu, Z., Xu, B., Cheng, B., Hu, X., Darbandi, M.: Intrusion detection systems in the cloud computing: a comprehensive and deep literature review. Concurr. Comput. Pract. Exp. **34**(4), 6646 (2022). https://doi.org/10.1002/cpe.6646
17. Ahmadi, S.: Systematic literature review on cloud computing security: threats and mitigation strategies. J. Inf. Secur. **15**, 148–167 (2024). https://doi.org/10.4236/jis.2024.152010
18. Preethi, M.: Intrusion detection and secure data storage in the cloud. Expert Syst. Appl. **242**, 122946 (2024). https://doi.org/10.1016/j.eswa.2023.122946
19. Nasim, S.S., Pranav, P., Dutta, S.: A systematic literature review on intrusion detection techniques in cloud computing. Discov. Comput. **28**(1), 107 (2025). https://doi.org/10.1007/s10791-025-09641-y
20. Bellouch, S., Aouad, M. Sihamand Zbakh: Smart intrusion detection system in cloud computing: a systematic review. In: Farhaoui, Y., Herawan, T., Imoize, A.L., Allaoui, A.E. (eds.) Intersection of Artificial Intelligence, Data Science, and Cutting-Edge Technologies: From Concepts to Applications in Smart Environment, pp. 41–47. Springer, ??? (2025). https://doi.org/10.1007/978-3-031-88304-0_6
21. Dong, H., Kotenko, I.: Cybersecurity in the ai era: analyzing the impact of machine learning on intrusion detection. Knowl. Inf. Syst. **67**(5), 3915–3966 (2025). https://doi.org/10.1007/s10115-025-02366-w
22. Nassif, A.B., Talib, M.A., Nasir, Q., Albadani, H., Dakalbab, F.M.: Machine learning for cloud security: a systematic review. IEEE Access **9**, 20717–20735 (2021). https://doi.org/10.1109/ACCESS.2021.3054129
23. Tawalbeh, L., Almomani, R., Daraghmeh, A.: Enhanced detection of distributed denial of service attacks in cloud platforms using machine learning. Clust. Comput. **29**(4), 200 (2026)
24. Vu, L., Nguyen, Q.U., Nguyen, D.N., Hoang, D.T., Dutkiewicz, E.: Deep generative learning models for cloud intrusion detection systems. IEEE Trans. Cybern. **53**(1), 565–577 (2023). https://doi.org/10.1109/TCYB.2022.3163811
25. Mohale, V.Z., Obagbuwa, I.C.: Evaluating machine learning-based intrusion detection systems with explainable ai: enhancing transparency and interpretability. Front. Comput. Sci. (2025). https://doi.org/10.3389/fcomp.2025.1520741
26. Abdallah, A.M., Saif Rashed ObaidAlkaabi, A., Bark Nasser DoumanAlameri, G., Rafique, S.H., Musa, N.S., Murugan, T.: Cloud network anomaly detection using machine and deep learning techniques- recent research advancements. IEEE Access **12**, 56749–56773 (2024). https://doi.org/10.1109/ACCESS.2024.3390844
27. Sruthi Mol, P., Sathish Kumar, N.: Efficient intrusion detection in cloud environments using optimized sparse and contractive autoencoders. Trans. Emerg. Telecommun. Technol. **37**(2), 70367 (2026)
28. Kitchenham, B., Brereton, O.P., Budgen, D., Turner, M., Bailey, J., Linkman, S.: Systematic literature reviews in software engineering – a systematic literature review. Inf. Softw. Technol. **51**(1), 7–15 (2009). https://doi.org/10.1016/j.infsof.2008.09.009
29. Belal, M.M., Sundaram, D.M.: Comprehensive review on intelligent security defences in cloud: taxonomy, security issues, ml/dl techniques, challenges and future trends. J. King Saud Univ. Comput. Inf. Sci. **34**(10), 9102–9131 (2022). https://doi.org/10.1016/j.jksuci.2022.08.035. (**(Part B)**)
30. Pimentel, E.L., Souza, C.A., Westphall, C.B.: Detecting attacks in fog and cloud computing environments using deep learning: a systematic literature review. Comput. Netw. **264**, 111269 (2025). https://doi.org/10.1016/j.comnet.2025.111269
31. Maodah, K.A.: Integrating machine learning and intrusion detection for improved cloud security: a comprehensive survey. Int. J. Cloud Appl. Comput. (IJCAC) **16**(1), 1–21 (2026). https://doi.org/10.4018/IJCAC.411871. Open Access
32. Velliyath, S., Kalaivani, D.: Adversarial challenges in AI-based intrusion detection systems for cloud environments: a comprehensive review. In: 2025 5th International Conference on Evolutionary Computing and Mobile Sustainable Networks (ICECMSN), pp. 553–560. IEEE, ??? (2025). https://doi.org/10.1109/ICECMSN68058.2025.11382855
33. Ma, X., Yang, L., Wang, H., Ling, C.: Model optimization in federated intrusion detection systems: a survey of aggregation, communication, and personalization. J. Cloud Comput. **15**(1), 88 (2026). https://doi.org/10.1186/s13677-026-00901-w
34. Buyuktanir, B., Altinkaya, Ş, Karatas Baydogmus, G., Yildiz, K.: Federated learning in intrusion detection: advancements, applications, and future directions. Clust. Comput. **28**(7), 473 (2025). https://doi.org/10.1007/s10586-025-05325-w
35. MITRE Corporation: ATT&CK matrix for enterprise - cloud. https://attack.mitre.org/matrices/enterprise/cloud/. Accessed: 2025-08-18 (2025)

36. Khan, M.U., Sherin, S., Iqbal, M.Z., Zahid, R.: Landscaping systematic mapping studies in software engineering: a tertiary study. J. Syst. Softw. **149**, 396–436 (2019). https://doi.org/10.1016/j.jss.2018.12.018
37. Maheswari, K.G., Siva, C., Priya, G.N.: An optimal cluster based intrusion detection system for defence against attack in web and cloud computing environments. Wireless Pers. Commun. **128**(3), 2011–2037 (2023). https://doi.org/10.1007/s11277-022-10030-7
38. Silambarasan, E., Suryawanshi, R., Reshma, S.: Enhanced cloud security: a novel intrusion detection system using arso algorithm and bi-lstm classifier. Int. J. Inf. Technol. **16**(6), 3837–3845 (2024). https://doi.org/10.1007/s41870-024-01887-x
39. Nagamani, S., Arivalagan, S., Senthil, M., Sudhakar, P.: Horse herd optimization with deep learning based intrusion detection in cloud computing environment. Int. J. Inf. Technol. **17**(1), 387–393 (2025). https://doi.org/10.1007/s41870-024-02199-w
40. Bajpai, S.A., Patankar, A.B.: Marine goal optimizer tuned deep bilstm-based self-configuring intrusion detection in cloud. J. Grid Comput. **22**(1), 24 (2024). https://doi.org/10.1007/s10723-023-09728-0
41. Jain, D.K., Ding, W., Kotecha, K.: Training fuzzy deep neural network with honey badger algorithm for intrusion detection in cloud environment. Int. J. Mach. Learn. Cybern. **14**(6), 2221–2237 (2023). https://doi.org/10.1007/s13042-022-01758-6
42. Ghosh, P., Alam, Z., Sharma, R.R., Phadikar, S.: An efficient sgm based ids in cloud environment. Computing **104**(3), 553–576 (2022). https://doi.org/10.1007/s00607-022-01059-4
43. Parkash, D., Mittal, S.: An enhanced secure framework using csa for cloud computing environments. In: Gupta, D., Khanna, A., Bhattacharyya, S., Hassanien, A.E., Anand, S., Jaiswal, A. (eds.) International Conference on Innovative Computing and Communications, pp. 349–356. Springer, Singapore (2023). https://doi.org/10.1007/978-981-19-2535-1_27
44. Sarkar, N., Keserwani, P.K., Govil, M.C.: A better and fast cloud intrusion detection system using improved squirrel search algorithm and modified deep belief network. Clust. Comput. **27**(2), 1699–1718 (2023). https://doi.org/10.1007/s10586-023-04037-3
45. Arunkumar, M., Kumar, K.A.: Gosvm: gannet optimization based support vector machine for malicious attack detection in cloud environment. Int. J. Inf. Technol. **15**(3), 1653–1660 (2023). https://doi.org/10.1007/s41870-023-01192-z
46. Bakro, M., Kumar, R.R., Husain, M., Ashraf, Z., Ali, A., Yaqoob, S.I., Ahmed, M.N., Parveen, N.: Building a cloud-ids by hybrid bio-inspired feature selection algorithms along with random forest model. IEEE Access **12**, 8846–8874 (2024). https://doi.org/10.1109/ACCESS.2024.3353055
47. Sanagana, D.P.R., Tummalachervu, C.K.: Securing cloud computing environment via optimal deep learning-based intrusion detection systems. In: 2024 Second International Conference on Data Science and Information System (ICDSIS), pp. 1–6 (2024). https://doi.org/10.1109/ICDSIS61070.2024.10594404
48. Eswari, G., Monica, G.K., Deepak, V., Sunil, K.M., Kumar, B.P.: Enhancing cloud storage security with intrusion detection system using cnn and grey wolf optimization algorithm. In: 2023 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS), pp. 557–563 (2023). https://doi.org/10.1109/ICSCDS56580.2023.10104643
49. Sreelatha, G., Babu, A.V., Midhunchakkaravarthy, D.: Improved security in cloud using sandpiper and extended equilibrium deep transfer learning based intrusion detection. Clust. Comput. **25**(5), 3129–3144 (2022). https://doi.org/10.1007/s10586-021-03516-9
50. JansiSophiaMary, C., Mahalakshmi, K.: Modelling of intrusion detection using sea horse optimization with machine learning model on cloud environment. Int. J. Inf. Technol. **16**(3), 1981–1988 (2024). https://doi.org/10.1007/s41870-023-01722-9
51. Ravindranath, V., Ramasamy, S., Somula, R., Sahoo, K.S., Gandomi, A.H.: Swarm intelligence based feature selection for intrusion and detection system in cloud infrastructure. In: 2020 IEEE Congress on Evolutionary Computation (CEC), pp. 1–6 (2020). https://doi.org/10.1109/CEC48606.2020.9185887
52. Radhika, R.: Intrusion detection system using improved pelican optimization algorithm-based support vector machine to secure data in cloud. In: 2024 International Conference on Data Science and Network Security (ICDSNS), pp. 1–7 (2024). https://doi.org/10.1109/ICDSNS62112.2024.10691077
53. Bai, V.S., Punithavalli, M.: Leveraging feature subset selection with deer hunting optimizer based deep learning for anomaly detection in secure cloud environment. Mult. Tools Appl. **83**(25), 65949–65966 (2024). https://doi.org/10.1007/s11042-024-18162-7
54. Jaber, A.N., Rehman, S.U.: Fcm–svm based intrusion detection system for cloud computing environment. Clust. Comput. **23**(4), 3221–3231 (2020). https://doi.org/10.1007/s10586-020-03082-6
55. Pandey, V.K., Shukla, R., Kumar, A., Singh, M.: A lightweight framework to secure iot devices with limited resources. Sci. Rep. **15**, 9825 (2025). https://doi.org/10.1038/s41598-025-09885-0
56. Kaur, A., Tyagi, R.K., Sinha, S., Vats, P.: Systems for detecting intruders in internet of things environments using the cloud: A survey. In: Emerging Trends in Industrial Engineering and Management, pp. 11–38. CRC Press, ??? (2026)
57. Manzoor, S., Ahmad, M., Alhadawi, H.S.: Intrusion detection system using ensemble machine learning in cloud environment. In: Al-Sharafi, M.A., Al-Emran, M., Al-Kabi, M.N., Shaalan, K. (eds.) Proceedings of the 2nd International Conference on Emerging Technologies and Intelligent Systems, pp. 513–522. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-25274-7_43
58. Tiwari, R.K., Murugappan, S.: Enhancing security in cloud computing and protocols using harris hawks optimizer with deep learning for intrusion detection. In: 2023 International Conference on Sustainable Communication Networks and Application (ICSCNA), pp. 170–175 (2023).https://doi.org/10.1109/ICSCNA58489.2023.10370663
59. Rana, P., Batra, I., Malik, A., Ra, I.-H., Lee, O.-S., Sanwar Hosen, A.S.M.: Efficacious novel intrusion detection system for cloud computing environment. IEEE Access **12**, 99223–99239 (2024). https://doi.org/10.1109/ACCESS.2024.3424528
60. Attou, H., Guezzaz, A., Benkirane, S., Azrour, M.: A sorting-ab approach for an enhanced ids in cloud environment. In: Farhaoui, Y., Rocha, A., Brahmia, Z., Bhushab, B. (eds.) Artif. Intell. Smart Environ., pp. 120–125. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-26254-8_18
61. Kim, Y., Park, G., Kim, H.K.: Domain knowledge free cloud-ids with lightweight embedding method. J. Cloud Comput. **13**(1), 20 (2024). https://doi.org/10.1186/s13677-024-00707-8
62. Joraviya, N., Gohil, B.N., Rao, U.P.: Dl-hids: deep learning-based host intrusion detection system using system calls-to-image for containerized cloud environment. J. Supercomput. **80**(9), 12218–12246 (2024). https://doi.org/10.1007/s11227-024-05895-3
63. Liu, M., Xue, Z., He, X.: A unified host-based intrusion detection framework using spark in cloud. In: 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), pp. 97–103 (2020). https://doi.org/10.1109/TrustCom50675.2020.00026
64. Gill, K.S., Saxena, S., Sharma, A.: Ncgtm: a noncooperative game-theoretic model to assist ids in cloud environment. IEEE Trans. Industr. Inf. **20**(3), 3124–3132 (2024). https://doi.org/10.1109/TII.2023.3300452
65. Ghosh, P., Sinha, S., Sharma, R.R., Phadikar, S.: An efficient ids in cloud environment using feature selection based on dm algorithm. J. Comput. Virol. Hack. Tech. **18**(3), 243–258 (2022). https://doi.org/10.1007/s11416-021-00410-1

66. Mitropoulou, K., Kokkinos, P., Soumplis, P., Varvarigos, E.: Anomaly detection in cloud computing using knowledge graph embedding and machine learning mechanisms. J. Grid Comput. **22**(1), 6 (2023). https://doi.org/10.1007/s10723-023-09727-1
67. Sethi, K., Kumar, R., Mohanty, D., Bera, P.: Robust adaptive cloud intrusion detection system using advanced deep reinforcement learning. In: Security, Privacy, and Applied Cryptography Engineering: 10th International Conference, SPACE 2020, Kolkata, India, December 17–21, 2020, Proceedings, pp. 66–85. Springer, Berlin, Heidelberg (2020). https://doi.org/10.1007/978-3-030-66626-2_4
68. Bakro, M., Kumar, R.R., Bisoy, S.K., Addas, M.O., Khamis, D.: Developing a cloud intrusion detection system with filter-based features selection techniques and svm classifier. In: Panda, S.K., Rout, R.R., Sadam, R.C., Rayanoothala, B.V.S., Li, K.-C., Buyya, R. (eds.) Computing, Communication and Learning, pp. 15–26. Springer (2022). https://doi.org/10.1007/978-3-031-21750-0_2
69. Gupta, R., Alam, T.: An efficient federated learning based intrusion detection system using ls2dnn with pbka based lightweight privacy preservation in cloud server. Mult. Tools Appl. **83**(15), 44685–44697 (2024). https://doi.org/10.1007/s11042-023-17401-7
70. Zhu, J., He, S., He, P., Liu, J., Lyu, M.R.: Loghub: A large collection of system log datasets for ai-driven log analytics. In: 2023 IEEE 34th International Symposium on Software Reliability Engineering (ISSRE), pp. 355–366 (2023). https://doi.org/10.1109/ISSRE59848.2023.00071
71. Ayed, M.A., Talhi, C.: Federated learning for anomaly-based intrusion detection. In: 2021 International Symposium on Networks, Computers and Communications (ISNCC), pp. 1–8 (2021). https://doi.org/10.1109/ISNCC52172.2021.9615816
72. Luo, H., Ji, C.: Cross-cloud data privacy protection: optimizing collaborative mechanisms of AI systems by integrating federated learning and LLMs. In: 2025 IEEE 7th International Conference on Communications, Information System and Computer Engineering (CISCE) (2025). https://doi.org/10.1109/cisce65916.2025.11065466
73. Tyagi, J., Sharma, M.: Enhancing privacy-preserving data mining in cloud-based data warehouses: a federated learning approach for secure multi-tenant environments. In: 2025 IEEE 10th International Conference on Smart Cloud (SmartCloud), pp. 8–13 (2025). https://doi.org/10.1109/SmartCloud66068.2025.00016
74. Bhingarkar, S., Revathi, K., Sekhar, C., Mewada, H.K.: An effective optimization enabled deep learning based malicious behaviour detection in cloud computing. Int. J. Intell. Robot. Appl. **13**(1), 575–588 (2023). https://doi.org/10.1007/s41315-022-00239-x
75. Jiang, G.: Artificial intelligence-based adaptive anomaly detection technology for iaas cloud virtual machines. J. Eng. Appl. Sci. **71**(1), 102 (2024). https://doi.org/10.1186/s44147-024-00439-1
76. Zuhair, M., Bhattacharya, P., Prasad, V.K., Barot, M., Modi, M.: Analysis of boosting mechanisms in cloud-based intrusion detection systems. In: 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), pp. 961–966 (2022). https://doi.org/10.1109/IC3I56241.2022.10072683
77. Parameswarappa, P., Shah, T., Lanke, G.R.: A machine learning-based approach for anomaly detection for secure cloud computing environments. In: 2023 International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), pp. 931–940 (2023). https://doi.org/10.1109/IDCIoT56793.2023.10053518
78. Chkirbene, Z., Hamila, R., Erbad, A., Kiranyaz, S., Al-Emadi, N., Hamdi, M.: Cooperative machine learning techniques for cloud intrusion detection. In: 2021 International Wireless Communications and Mobile Computing (IWCMC), pp. 837–842 (2021). https://doi.org/10.1109/IWCMC51323.2021.9498809
79. Shahzad, F., Mannan, A., Javed, A.R., Almadhor, A.S., Baker, T., Al-Jumeily Obe, D.: Cloud-based multiclass anomaly detection and categorization using ensemble learning. J. Cloud Comput. **11**(1), 74 (2022). https://doi.org/10.1186/s13677-022-00329-y
80. Al-Ghuwairi, A.-R., Sharrab, Y., Al-Fraihat, D., AlElaimat, M., Alsarhan, A., Algarni, A.: Intrusion detection in cloud computing based on time series anomalies utilizing machine learning. J. Cloud Comput. **12**(1), 17 (2023). https://doi.org/10.1186/s13677-023-00491-x
81. Xin, R., Liu, H., Chen, P., Zhao, Z.: Robust and accurate performance anomaly detection and prediction for cloud applications: a novel ensemble learning-based framework. J. Cloud Comput. (2023). https://doi.org/10.1186/s13677-022-00383-6
82. Althoubi, A., Peyravi, H.: A hybrid intrusion detection system leveraging xgboost and rnns for enhanced anomaly detection in cloud data centers. In: 2023 International Conference on Computational Science and Computational Intelligence (CSCI), pp. 1039–1046. IEEE Computer Society, Los Alamitos, CA, USA (2023). https://doi.org/10.1109/CSCI62032.2023.00172
83. Sethi, K., Kumar, R., Prajapati, N., Bera, P.: Deep reinforcement learning based intrusion detection system for cloud infrastructure. In: 2020 International Conference on COMmunication Systems & NETworkS (COMSNETS), pp. 1–6 (2020). https://doi.org/10.1109/COMSNETS48256.2020.9027452
84. Attou, H., Guezzaz, A., Benkirane, S., Azrour, M., Farhaoui, Y.: Cloud-based intrusion detection approach using machine learning techniques. Big Data Min. Analy. **6**(3), 311–320 (2023)
85. Holla, H., Polepalli, S.R., Sasikumar, A.A.: Adversarial threats to cloud ids: robust defense with adversarial training and feature selection. IEEE Access **13**, 84992–85003 (2025). https://doi.org/10.1109/ACCESS.2025.3567038
86. Patel, A., Pandey, P., Ragothaman, H., Molleti, R., Tanikonda, A.: Securing cloud ai workloads: Protecting generative ai models from adversarial attacks. In: 2025 IEEE 4th International Conference on AI in Cybersecurity (ICAIC), pp. 1–7 (2025). https://doi.org/10.1109/ICAIC63015.2025.10848877
87. Peng, Y., Fu, G., Luo, Y., Hu, J., Li, B., Yan, Q.: Detecting adversarial examples for network intrusion detection system with gan. In: 2020 IEEE 11th International Conference on Software Engineering and Service Science (ICSESS), pp. 6–10 (2020). https://doi.org/10.1109/ICSESS49938.2020.9237728
88. Ganesan, A., Sarac, K.: Mitigating evasion attacks on machine learning based nids systems in sdn. In: 2021 IEEE 7th International Conference on Network Softwarization (NetSoft), pp. 268–272 (2021). https://doi.org/10.1109/NetSoft51509.2021.9492526
89. Hizal, S., ÇAVUŞOĞLU, E., AKGÜN, D.: A new deep learning based intrusion detection system for cloud security. In: 2021 3rd International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA), pp. 1–4 (2021). https://doi.org/10.1109/HORA52670.2021.9461285
90. Geetha, T.V., Deepa, A.J., Linda, M.M.: Deep learning method for efficient cloud ids utilizing combined behavior and flow-based features. Appl. Intell. **54**(8), 6738–6759 (2024). https://doi.org/10.1007/s10489-024-05505-y
91. Venkateshwarlu, V., Ranjith, D., Raju, A.: Lrdadf: an ai enabled framework for detecting low-rate ddos attacks in cloud computing environments. In: 2023 Fifth International Conference on Electrical, Computer and Communication Technologies (ICECCT), pp. 1–8 (2023). https://doi.org/10.1109/ICECCT56650.2023.10179834
92. Srilatha, D., Shyam, G.K.: Cloud-based intrusion detection using kernel fuzzy clustering and optimal type-2 fuzzy neural network. Clust. Comput. **24**(3), 2657–2672 (2021). https://doi.org/10.1007/s10586-021-03281-9
93. Bakro, M., Kumar, R.R., Alabrah, A., Ashraf, Z., Ahmed, M.N., Shameem, M., Abdelsalam, A.: An improved design for

a cloud intrusion detection system using hybrid features selection approach with ml classifier. IEEE Access **11**, 64228–64247 (2023). https://doi.org/10.1109/ACCESS.2023.3289405
94. Singh, P., Ranga, V.: Attack and intrusion detection in cloud computing using an ensemble learning approach. Int. J. Inf. Technol. **13**(2), 5650571 (2021). https://doi.org/10.1007/s41870-020-00583-w
95. Daoud, M.A., Dahmani, Y., Ammar, S., Ouared, A.: Classifying of intrusion detection system configurations using machine learning techniques. In: 2021 International Conference on Information Systems and Advanced Technologies (ICISAT), pp. 1–6 (2021). https://doi.org/10.1109/ICISAT54145.2021.9678473
96. Khan, M., Haroon, M.: Detecting network intrusion in cloud environment through ensemble learning and feature selection approach. SN Comput. Sci. **5**(1), 9 (2023). https://doi.org/10.1007/s42979-023-02390-z
97. Shukla, A.K., Sharma, A.: Cloud base intrusion detection system using convolutional and supervised machine learning. In: 2023 6th International Conference on Information Systems and Computer Networks (ISCON), pp. 1–5 (2023). https://doi.org/10.1109/ISCON57294.2023.10112007
98. Jiang, W., Cheng, Y., Li, Y., Zhao, M.: Cloud environment intrusion detection based on gan and lightgbm. In: 2023 IEEE 3rd International Conference on Electronic Technology, Communication and Information (ICETCI), pp. 1439–1443 (2023). https://doi.org/10.1109/ICETCI57876.2023.10177025
99. R, A., Das, U., AK, D.R., Balachandar, N.: Cloud-based intrusion detection system using various machine learning techniques. In: 2024 5th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV), pp. 413–420 (2024). https://doi.org/10.1109/ICICV62344.2024.00071
100. Kassimi, D., Kazar, O., Barka, E., Merizig, A., Houhamdi, Z., Athamena, B., Zaoui, M.: A new approach based on a multi–agent system for ids in cloud computing. In: 2022 Ninth International Conference on Software Defined Systems (SDS), pp. 1–8 (2022). https://doi.org/10.1109/SDS57574.2022.10062930
101. Fugkeaw, S., Moolkaew, N., Wiwattanapornpanit, T., Saengsena, T., Sanchol, P.: A resilient cloud-based ddos attack detection and prevention system. In: 2023 20th International Joint Conference on Computer Science and Software Engineering (JCSSE), pp. 184–189 (2023). https://doi.org/10.1109/JCSSE58229.2023.10202023
102. Tripathy, D., Gohil, R., Halabi, T.: Detecting sql injection attacks in cloud saas using machine learning. In: 2020 IEEE 6th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing, (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS), pp. 145–150 (2020). https://doi.org/10.1109/BigDataSecurity-HPSC-IDS49724.2020.00035
103. Onyema, E.M., Dalal, S., Romero, C.A.T., Seth, B., Young, P., Wajid, M.A.: Design of intrusion detection system based on cyborg intelligence for security of cloud network traffic of smart cities. J. Cloud Comput. **11**(1), 20 (2022). https://doi.org/10.1186/s13677-022-00305-6
104. Veena, C., Ramalakshmi, S., Bhoopathy, V., Bhosale, M.D., Magadum, C.G., K., A.S.: Effective intrusion detection and classification using fuzzy rule based classifier in cloud environment. In: 2022 International Conference on Automation, Computing and Renewable Systems (ICACRS), pp. 497–502 (2022). https://doi.org/10.1109/ICACRS55517.2022.10028988
105. Wang, X.: Fast localization model of network intrusion detection system for enterprises using cloud computing environment. Mobile Networks Appl. **28**(6), 2191–2203 (2023). https://doi.org/10.1007/s11036-023-02176-w
106. Krishnaveni, S., Sivamohan, S., Sridhar, S.S., Prabakaran, S.: Efficient feature selection and classification through ensemble method for network intrusion detection on cloud computing. Clust. Comput. **24**(3), 1761–1779 (2021). https://doi.org/10.1007/s10586-020-03222-y
107. Kandakatla, R., Rajakumari, K.: Hyperparameter-tuned machine learning model for the ids security in cloud environment. In: 2024 International Conference on Intelligent Algorithms for Computational Intelligence Systems (IACIS), pp. 1–8 (2024). https://doi.org/10.1109/IACIS61494.2024.10721918
108. Kumar, A., Sharma, I.: Methodology for safeguarding cloud server from web application attacks. In: 2023 International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE), pp. 1–6 (2023). https://doi.org/10.1109/RMKMATE59243.2023.10369725
109. Sharma, D., Tiwari, D., Singh, V., Pandey, P., Gupta, V.K.: Cloud-based occlusion aware intrusion detection system. In: Jain, R., Travieso, C.M., Kumar, S. (eds.) Cybersecurity and Evolutionary Data Engineering, pp. 69–78. Springer, Singapore (2023). https://doi.org/10.1007/978-981-99-5080-5_7
110. Jisna, P., Jarin, T., Praveen, P.N.: Advanced intrusion detection using deep learning-lstm network on cloud environment. In: 2021 Fourth International Conference on Microelectronics, Signals & Systems (ICMSS), pp. 1–6 (2021). https://doi.org/10.1109/ICMSS53060.2021.9673607
111. Kairouz, P., McMahan, H.B., Avent, B., et al.: Advances and open problems in federated learning. Found. Trends Mach. Learn. **14**(1–2), 1–210 (2021). https://doi.org/10.1561/2200000083
112. Al-Mazrawe, A.: Anomaly detection in cloud network: a review. Bio-Conferences (ISCKU2024), p. 16. (2024)
113. Alotaibi, A., Rassam, M.A.: Adversarial machine learning attacks against intrusion detection systems: a survey on strategies and defense. Fut. Internet **15**(2), 62 (2023). https://doi.org/10.3390/fi15020062
114. Goodman, D.: Transferability of adversarial examples to attack cloud-based image classifier service. In: DEFCON 2019, pp. 1–6 (2020). https://doi.org/10.48550/arXiv.2001.03460
115. Srinivasulu, B., Anil, D., Sowmya M, L., Katuri, L., Swapna Rani, K.: Cloud security in a multicloud world: best practices for protecting data and applications across multiple platforms. In: 2024 4th International Conference on Mobile Networks and Wireless Communications (ICMNWC), pp. 01–06 (2024). https://doi.org/10.1109/ICMNWC63764.2024.10872028
116. Chaudhary, A., M, N.S., Tiwaskar, S.A., Vijayalakshmi, V.J., Sharma, S., S, S.: Deep reinforcement learning for accurate anomaly detection in multi-cloud environment. In: 2024 Global Conference on Communications and Information Technologies (GCCIT), pp. 1–8 (2024). https://doi.org/10.1109/GCCIT63234.2024.10862913
117. J, S., Kanagasabapathi, K., Mahajan, K., Ahamad, S., Soumya, E., Barthwal, S.: Ai-enhanced multi-cloud security management: ensuring robust cybersecurity in hybrid cloud environments. In: 2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), pp. 1–6 (2023). https://doi.org/10.1109/ICSES60034.2023.10465550
118. Nizamudeen, S.M.T.: Intelligent intrusion detection framework for multi-clouds-iot environment using swarm-based deep learning classifier. J. Cloud Comput. **12**(1), 134 (2023). https://doi.org/10.1186/s13677-023-00509-4
119. Alazab, M., Awajan, A., Obeidat, A., Faruqui, N., Bere, A., Ali, S., Wei, W.: Adaptive protocols for hypervisor security in cloud infrastructure using federated learning-based anomaly detection. Eng. Appl. Artif. Intell. **152**, 110750 (2025). https://doi.org/10.1016/j.engappai.2025.110750
120. Anumukonda, N.S.K., Yadav, R.K., Raghava, N.S.: Hypervisor based intrusion detection using enhanced radial basis neural network on cloud environment. In: 2024 IEEE International

Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), vol. 2, pp. 1–6 (2024). https://doi.org/10.1109/IATMSI60426.2024.10503284
121. Kumar, S., Kumar, P., Kumar, J., Singh, V., Yadav, A.S.: Securing cloud-based systems: ddos attack mitigation using hypervisor-intrusion detection approach. Proced. Comput. Sci. **259**, 1366–1375 (2025). https://doi.org/10.1016/j.procs.2025.04.091
122. Qaffas, A.A.: Reinforcement learning driven self-adaptation in hypervisor-based cloud intrusion detection systems (rldac-ids). Int. J. Adv. Comput. Sci. Appl. (2024). https://doi.org/10.14569/IJACSA.2024.0150745
123. Alharkan, I., Alenezi, M.: A survey of hybrid and multi-cloud security: challenges and directions. IEEE Access **10**, 113140–113159 (2022). https://doi.org/10.1109/ACCESS.2022.3204138
124. Khan, M.A., Madani, S.A., Imran, M., Jan, M.A.: Survey on virtualization technologies in cloud computing. IEEE Access **10**, 53056–53078 (2022). https://doi.org/10.1109/ACCESS.2022.3174817
125. Mickens, J., Radway, S., Netravali, R.: Guillotine: hypervisors for isolating malicious ais. In: HotOS '25: Proceedings of the 2025 Workshop on Hot Topics in Operating Systems, pp. 18–26. Association for Computing Machinery, New York, NY, USA (2025). https://doi.org/10.1145/3713082.3730391
126. Balbix Inc.: Understanding MITRE ATT&CK for cloud. Accessed: 20-Nov-2025 (2023). https://www.balbix.com/insights/mitre-attck-for-cloud/
127. Kavi, B.: MITRE cloud matrix: a comprehensive overview. Accessed: 20-Nov-2025 (2023). https://medium.com/@kavib/mitre-cloud-matrix-a-comprehensive-overview-c204fac41fa
128. Zhao, T., Gasiba, T., Lechner, U., Pinto-Albuquerque, M.: Raising awareness about cloud security in industry through a board game. Information (2021). https://doi.org/10.3390/info12110482
129. Zhao, T., Lechner, U., Pinto-Albuquerque, M., Ata, E., Gasiba, T.E.: Cats: a serious game in industry towards stronger cloud security. In: Ubiquit. Secur., pp. 64–82. Springer (2023). https://doi.org/10.1007/978-981-99-0272-9_5
130. Xiong, W., Legrand, E., Åberg, O., Lagerström, R.: Cyber security threat modeling based on the mitre enterprise att&ck matrix. Softw. Syst. Model. **21**(1), 157–177 (2025). https://doi.org/10.1007/s10270-021-00898-7
131. Maccarone, L.T., Buede, D.M., Bowman, S.T., Ambrozewicz, P., Burdick, C.D., Grady, C., Wen, J.S.X.: Identifying adversarial cyber-activity in operational technology environments using bayesian networks. IEEE Trans. Inf. Forensics Secur. **20**, 10173–10188 (2025). https://doi.org/10.1109/TIFS.2025.3607241
132. Corporation, M.: MITRE ATLAS: adversarial threat landscape for artificial-intelligence systems. Accessed: Nov. 27, 2025 (2025). https://atlas.mitre.org/

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.